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ChatWise: Adaptive Support for Students' Self-Regulated Learning in Generative AI–Mediated Environments

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Abstract

Large Language Models (LLMs) such as ChatGPT have rapidly become part of students' everyday learning practices in higher education, particularly within STEM disciplines. While these tools can support programming, problem-solving, and writing, their convenience also introduces risks of overreliance, uncritical acceptance of automatically generated outputs, and cognitive offloading that may undermine self-regulated learning (SRL) — the ability to plan, monitor, and reflect on one's learning. Despite growing awareness of these challenges, few existing AI tools are explicitly designed to foster SRL during students' everyday interactions with commercial LLMs.

This thesis addresses this gap by introducing *ChatWise*, a browser extension developed to strengthen students' SRL when using generative AI tools. Instead of restricting the AI system itself, ChatWise focuses on helping students engage with ChatGPT more reflectively and strategically. The tool provides adaptive, metacognitive feedback in real time, classifying each prompt using AI-based real-time classification grounded in SRL theory to the SRL strategies elaboration, effort regulation, organisation, rehearsal, help seeking, and critical thinking. Through immediate visual and textual feedback, ChatWise encourages students to reflect on their prompting behaviour and rephrase prompts in ways that better support learning.

Following a Design-Based Research (DBR) approach, ChatWise was iteratively designed and evaluated through a within-subject study conducted in an authentic learning setting, involving 17 STEM undergraduates, of whom 9 completed both study phases. Quantitative analyses of prompt logs and survey data showed an increase in high-quality, SRL-aligned prompts when students used ChatWise, with substantial variation in how individual students' prompting strategies developed over time. Qualitative findings revealed increased reflection, greater awareness of productive prompting strategies, and improved engagement with learning tasks. Participants described the adaptive SRL feedback as motivating and helpful for increased understanding of how to interact with ChatGPT more intentionally for improved learning in STEM higher education.

These first results suggest that adaptive metacognitive feedback can foster more reflective and self-regulated engagement when interacting with generative AI during studies. This work contributes both practical and theoretical insights into how educational AI tools can promote SRL and metacognitive awareness. The thesis concludes with design recommendations for supporting reflective, sustainable, student-centred use of generative AI in

higher education.

Keywords

Self-Regulated Learning, Generative AI, ChatGPT, Adaptive Feedback, Design-Based Research, Higher Education, Metacognition, Prompting Strategies

Sammanfattning

Stora språkmodeller som ChatGPT har snabbt blivit en integrerad del av studenters vardagliga lärande i högre utbildning, särskilt inom STEM-ämnen. Dessa verktyg kan stödja programmering, problemlösning och skrivande, men deras lättillgänglighet innebär samtidigt risker: överberoende, okritisk tillit till AI-genererade svar och en form av kognitiv avlastning som kan hämma självreglerat lärande (SRL) – förmågan att planera, följa upp och reflektera över sitt eget lärande.Trots en växande medvetenhet om dessa utmaningar finns det fortfarande få AI-verktyg som är specifikt utformade för att stärka studenters SRL i interaktionen med generativ AI.

Detta examensarbete adresserar detta genom att introducera ChatWise, ett webbläsartillägg utvecklat för att stödja studenters SRL vid användning av kommersiella AI-verktyg som ChatGPT. I stället för att begränsa själva AI-systemet fokuserar ChatWise på att hjälpa studenter att interagera med ChatGPT på ett mer reflekterande och strategiskt sätt i sitt lärande. Verktyget ger flexibel och metakognitiv återkoppling i realtid genom AI-baserad klassificering av varje prompt enligt SRL-teori, där promptarna kategoriseras utifrån strategierna elaborering, ansträngningsreglering, organisering, repetition, hjälpsökande och kritiskt tänkande. Genom omedelbar visuell och textuell återkoppling uppmuntrar ChatWise studenter att reflektera över sina prompts och formulera dem på sätt som bättre stödjer deras lärande.

Utvecklingen av ChatWise följde en Design-Based Research (DBR)-ansats och utvärderades genom en within-subject-studie genomförd i ett sammanhang där studenterna vanligtvis studerar, med 17 STEM-studenter, varav 9 slutförde båda studieomgångarna. Kvantitativa analyser av promptloggar och enkäter visade en ökning av högkvalitativa prompts som ligger i linje med SRL-strategier vid användning av ChatWise. De kvalitativa resultaten visade ökad reflektion, större medvetenhet om effektiva promptstrategier och förbättrat engagemang i studenternas lärande. Deltagarna beskrev feedbacken från ChatWise som motiverande och hjälpsam för att förstå hur de kan interagera med ChatGPT på ett mer medvetet sätt.

Resultaten visar att adaptiv metakognitiv feedback kan främja ett mer reflekterande och självreglerat användande av generativ AI. Arbetet bidrar med både praktiska och teoretiska insikter kring hur utbildningsorienterade AI-verktyg kan utformas för att stödja självreglering och metakognitivt lärande. Uppsatsen avslutas med designrekommendationer för hur generativ AI kan användas på ett reflekterande, hållbart och studentcentrerat sätt i högre utbildning.

Nyckelord

Självreglerat lärande, Generativ AI, ChatGPT, Adaptiv återkoppling, Design-baserad forskning, Högre utbildning, Metakognition, Promptstrategier

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Chapter 1

Introduction

Large Language Models (LLMs) such as ChatGPT have rapidly become embedded in students' everyday learning practices in higher education [1, 2, 3]. In STEM disciplines in particular, students rely on these tools to generate code, debug, summarise theoretical concepts, and solve complex problems [4, 5, 6]. While generative AI can improve efficiency and accessibility, research shows that students often use LLMs to obtain quick answers rather than to deepen their understanding [7, 8, 9]. Such behaviour risks contributing to cognitive offloading [10] and what has been described as "metacognitive laziness," where learners bypass essential processes of evaluation, reflection, and strategic thinking [11].

Self-Regulated Learning (SRL)—learners' ability to plan, monitor, and reflect on their learning—is strongly associated with academic success across higher education [12, 13, 14]. SRL involves strategies such as goal setting, elaboration, critical thinking, effort regulation, and self-monitoring [15, 16]. However, current uses of LLMs seldom encourage these behaviours. Instead, students often rely on generative AI tools to complete tasks rather than to support meaningful learning processes [17, 18, 6]. This tension between efficiency and reflective engagement is becoming increasingly apparent in recent empirical findings.

A recent meta-analysis further demonstrates that although ChatGPT significantly improves learning performance relative to comparable student groups learning without ChatGPT, its impact on learning perception and higher-order thinking skills, such as problem-solving and critical thinking, remains moderate and highly dependent on *how* it is used [8]. Students who engage with LLMs superficially, accept model outputs uncritically, or delegate cognitive work to the system benefit far less in terms of deeper

understanding. These findings highlight that generative AI does not inherently promote metacognition or SRL, its educational value depends on the learner's approach and intention.

Despite these concerns, existing AI tools rarely provide guidance that helps university students use LLMs strategically during authentic study tasks. Most available systems are either designed for general audiences or targeted toward K–12 learners, offering limited support for SRL in higher education [19, 20, 17]. Popular tools such as ChatGPT provide content support but do not scaffold processes like goal-setting, monitoring, or reflection, and new features such as Study Mode do not explicitly target SRL strategies [21]. Studies further show that university students generally adopt LLMs for productivity rather than metacognitive development [22, 23]. This leaves a critical pedagogical gap: students extensively use generative AI tools, but often without structured support that helps them align their prompting strategies with evidence-based SRL practices.

This thesis addresses this gap by developing and evaluating ChatWise, a browser extension designed to foster SRL in real time during students' interactions with ChatGPT. Rather than restricting access, ChatWise supports metacognitive engagement by analysing students' prompts and providing adaptive feedback grounded in SRL theory. Through this approach, the aim is to help students move from passive consumption toward more intentional, reflective, and self-regulated use of generative AI in their studies.

1.1 Problem and Purpose

Although LLMs such as ChatGPT are now widely used by university students, the interaction patterns they encourage rarely align with established SRL strategies. Students tend to use these tools for task completion rather than for planning, monitoring, or reflecting on their learning processes [9, 7]. Existing AI tools offer limited support for helping learners adopt more intentional and metacognitively oriented approaches, particularly in higher education contexts [19, 17].

The practical problem addressed in this thesis is the absence of real-time SRL-oriented support for students when working with commercial LLM platforms such as ChatGPT. The scientific problem concerns how adaptive feedback mechanisms can be designed to encourage strategies such as critical thinking, elaboration, and reflection during AI-assisted learning [15, 17]. In this thesis, ChatWise is explicitly designed to increase the use of elaboration, organisation, critical thinking, and effort regulation strategies, while reducing

passive help-seeking and shallow rehearsal behaviours.

From an engineering perspective, this project involves the design and implementation of a system (ChatWise) that analyses students' interactions, namely the content of the prompts, with ChatGPT and providing adaptive, real-time textual feedback intended to promote SRL-supportive prompting behaviour. From a scientific perspective, the project explores how such feedback influences students' prompting behaviour when interacting with commercial LLMs and perceptions of their own learning regulation when using ChatWise.

The purpose of this thesis is therefore to design, implement, and evaluate ChatWise as a tool for supporting SRL in the context of generative AI use in higher education. The aim is to contribute to ongoing research on metacognitive support in AI-mediated learning environments and to inform the development of educational technologies that promote reflective, autonomous, and responsible engagement with LLMs, as well as to advance SRL-adaptive support and broader discussions on how AI can foster metacognitive and self-regulatory skills in higher education.

Ethical and social aspects are also considered. The use of AI in education raises questions about fairness, over-reliance, and responsible learning practices. This project seeks to encourage mindful and reflective use of AI that supports students' autonomy rather than replacing it.

1.2 Research Questions and Contributions

This thesis addresses the following research questions:

- **RQ1:** How do university students currently use ChatGPT to support their studies, and to what extent do these uses reflect SRL strategies?
- **RQ2:** How can adaptive support be designed within ChatGPT-based tools to scaffold students' planning, monitoring, and reflection processes?
- **RQ3:** How does the use of ChatWise influence students' SRL-related prompting behaviour during AI-assisted study tasks?

Based on these questions, the thesis makes the following contributions:

- An implementation of ChatWise, a browser extension that delivers adaptive, real-time feedback to support reflective prompting with ChatGPT.

- Empirical evidence that ChatWise increases students' awareness of effective prompting behaviours and supports SRL-related engagement.
- Design recommendations for future educational AI tools that balance adaptivity and learner autonomy.

1.3 Research Methodology

A Design-Based Research (DBR) approach was adopted, combining iterative design, implementation, and evaluation of ChatWise in authentic educational settings. DBR was chosen because it allows practical problem solving while generating theoretical insights about learning and technology use [24, 25].

The study follows a pragmatic research paradigm, focusing on generating actionable insights through iterative cycles of design, implementation, and evaluation. This approach assumes that knowledge is best developed through the interaction between theory and practice, where educational tools are tested and refined in authentic contexts.

Data collection included prompt logs, observations, and pre/post surveys assessing students' SRL strategies and perceptions of the tool. The methodology integrates both quantitative and qualitative analyses to explore how adaptive feedback influences behaviour and perception.

Alternative methodologies, such as controlled experiments or purely qualitative case studies, were considered less suitable since they would not capture the iterative and context-sensitive nature of developing and testing an interactive learning tool in authentic settings.

1.4 Delimitations

The thesis focuses on the design, implementation, and evaluation of ChatWise as a prototype that supports students' self-regulated learning when using generative AI. The project aims to explore behavioural and perceptual aspects of AI-assisted learning rather than long-term academic outcomes or system performance metrics.

The scope of this thesis is limited in several ways:

- The study focuses on STEM university students and may not generalize to other disciplines or educational levels.
- The evaluation captures short-term changes in prompting behaviours and perceptions, not long-term effects on learning outcomes.

- ChatWise was implemented specifically for ChatGPT and not integrated with other LLM-based systems.
- The study examines reflective and behavioural aspects of learning rather than assessing content accuracy or knowledge gain.

These delimitations were made to ensure feasibility within the scope of a master's thesis while maintaining a clear and focused contribution. Although the results are not intended to be statistically generalizable, they offer valuable insights into the design and use of adaptive feedback mechanisms in higher education learning environments.

1.5 Structure of the Thesis

Chapter 2 presents the theoretical background and related work on student use of generative AI, SRL, and existing learning tools. Chapter 3 outlines the research design and methodology. Chapter 4 describes the implementation and iterative design of ChatWise. Chapter 5 reports the empirical findings, followed by discussion and conclusions in Chapter 6.

Chapter 2

Background

2.1 Student Use of Generative AI in STEM Higher Education

The release of LLMs such as ChatGPT in 2022 has brought about significant changes in higher education, particularly within STEM disciplines, where students increasingly use these tools to support programming, problem-solving, and scientific writing [3]. ChatGPT quickly became a popular tool among university students in STEM fields around the world, offering the ability to write code, explain theoretical concepts, summarize content, and answer questions across technical subjects [3, 2].

Recent reports confirm this trend. For example, the recent *Anthropic Education Report* [1] found that a majority of university students, especially those in STEM disciplines such as computer science and engineering, have integrated LLMs such as Claude and ChatGPT into their everyday study routines. Notably, STEM students were found to use LLMs more frequently than peers in other fields, particularly for higher-order cognitive tasks such as debugging, conceptual reasoning, and complex problem-solving. This shows that, as institutions race to adapt, students have already normalized the use of LLMs as part of their academic workflow, as reported in the Anthropic Education Report [1].

While the adoption of generative AI tools is global, several studies confirm that student awareness and usage of ChatGPT are particularly widespread across higher education [26, 27]. For instance, a study conducted among Chinese university students established high awareness rates and strong intentions to adopt the tool, with perceived usefulness, ease of use, and trust emerging as key drivers of adoption [26]. Similarly, in Malaysia, students

reported being not only ready to use ChatGPT but also finding it highly useful for academic tasks such as completing assignments and searching for learning materials [27].

Comparable findings have been observed in U.S. higher education. A recent study involving STEM undergraduates across U.S. institutions found widespread adoption of ChatGPT and similar LLMs, with students emphasizing efficiency, accessibility, and writing support as key benefits for study-related tasks [5]. Likewise, a European study among engineering students revealed similar patterns, showing frequent use of ChatGPT for coding, conceptual explanations, and exam preparation [4]. Together, these results demonstrate that awareness and adoption of generative AI tools are widespread across cultural and educational contexts, highlighting a global shift in how students approach learning in STEM higher education.

Frequency of use is another important aspect. In a Peruvian study involving nearly 500 university students, intention to use ChatGPT on a regular basis was among the strongest predictors of positive attitudes toward it, along with responsible use and acceptance. Students who had a habit of verifying the information provided by ChatGPT were more likely to report a positive experience [28].

Prior research has documented a wide range of academic uses of ChatGPT among university students. Empirical studies show that students rely on generative AI for composing essays, debugging code, generating study summaries, and exploring alternative solutions to complex tasks [29, 2, 30]. Experimental research has further demonstrated that ChatGPT can enhance students' performance in creative and complex problem-solving tasks, improving their sense of self-efficacy and the quality of their output [31].

However, this convenience and efficiency are not without constraints. One concern relates to the notion of *cognitive offloading* — the delegation of mental processes such as memory or problem-solving to external systems [10] — and, more specifically, the phenomenon of "metacognitive laziness," where students become overly reliant on AI tools and skip important cognitive processes like reflection, evaluation, or revision [11]. These processes are closely linked to poorer learning outcomes, as skipping reflection, evaluation, and revision reduces opportunities for deeper understanding and self-regulated learning. This issue is especially relevant when students accept ChatGPT (or similar tools) outputs as final without questioning its accuracy, validity, or logic [9].

Furthermore, although ChatGPT can effectively support short-term task completion and create a perceived sense of learning ease and efficiency, it

does not consistently ensure enhanced intrinsic motivation or deeper learning outcomes. Research has for example, shown that students often rely on ChatGPT for quick solutions and surface-level understanding rather than sustained engagement or conceptual mastery [7].

Similarly, systematic reviews emphasize the importance of *structured frameworks*, i.e., pedagogical and metacognitive scaffolds that guide students in using LLMs purposefully, such as reflective prompts, step-by-step reasoning supports, and task-specific guidance [18]. Without such frameworks, the use of generative AI in higher education risks reinforcing passive learning habits rather than fostering active, self-regulated engagement.

Overall, these findings reveal a clear pedagogical guidance, i.e., while students are actively using LLMs, they often do so without awareness and guidance that supports meaningful engagement, understanding and ultimately learning. Without frameworks that encourage intentional and reflective use, students risk using LLMs to bypass learning rather than deepen it. This gap underscores the need for strategies that help students align their use of LLMs with the effective use of self-regulated learning strategy skills.

2.2 Self-Regulated Learning in STEM Education

Self-Regulated Learning (SRL) is a central theoretical framework for understanding how students plan, monitor, and reflect on their learning in higher education [14]. Students who regularly employ SRL strategies exhibit higher academic performance, higher motivation, and greater retention of learning materials [12, 32]. SRL refers to a process through which learners actively plan, monitor, and regulate their cognitive, motivational, and behavioural processes to achieve academic goals [13]. A foundational definition describes SRL as a goal-oriented activity in which learners make deliberate choices to guide and improve their learning performance [12]. This cyclical process typically includes phases of forethought, performance, and self-reflection, where learners evaluate and adapt their strategies to improve performance over time [12].

SRL is commonly operationalised through specific cognitive, metacognitive, and motivational strategies that learners use to regulate their learning. Several SRL models have been developed to elaborate on these processes, including those by Zimmerman, Pintrich, Winne and Hadwin, and Boekaerts; these frameworks collectively point to self-initiation, strategic action, and

adaptability as key characteristics of self-regulated learners [15].

To identify SRL in practice, researchers often focus on specific strategies. Common SRL strategies include elaboration (i.e., linking new knowledge with prior knowledge), critical thinking (i.e., evaluating arguments and evidence), rehearsal (repetition of material), organisation (i.e.,structuring information), time and study environment management, effort regulation, and help-seeking [16]. Evidence from empirical research shows that high-achieving students use elaboration and critical thinking strategies consistently, while lower-performing students tend to rely more on rehearsal and help-seeking behaviours [16].

In STEM and programming education, SRL plays a particularly important role due to high cognitive load, abstraction, and complex problem-solving demands. In programming education, a field known for high dropout rates and cognitive complexity [33, 34], SRL has been found to be a crucial factor for success. Studies indicate that students who self-regulate well cope better with cognitive load, debug more effectively, and store programming knowledge better [35, 32].

The increasing use of LLMs in STEM education introduces new challenges and opportunities for students' self-regulated learning. Recent research, including a study by Viberg *et al.* [6], highlights how LLMs are becoming increasingly used by students in programming education. These tools can support learning by providing explanations and feedback, but also raise concerns about cognitive offloading, where students may rely on AI outputs rather than engaging in effort regulation and critical thinking. This shift places even greater importance on fostering students' SRL knowledge and skills in digital environments [6].

Despite its theoretical importance, SRL is difficult to capture empirically, and existing research faces several methodological limitations. One major limitation is the reliance on self-report instruments, which may not capture the full complexity of SRL as it unfolds in real time [36]. To address this, recent studies have explored the use of multimodal data such as eye tracking, log data, and think-aloud protocols to obtain more objective and fine-grained insights into learners' self-regulatory behaviours [37]. These approaches reflect a growing recognition that SRL is best understood as a dynamic process that occurs during learning itself, rather than something students can accurately describe retrospectively [36]. By capturing learners' actions and decisions in context, such methods provide a more nuanced view of how self-regulation actually takes place in technology-rich learning environments [37].

Another research gap concerns the interaction between SRL and AI-

mediated learning environments. Although tools like adaptive learning systems and LLMs (e.g., ChatGPT or Claude) offer quick feedback and support, there is also the risk that they may unintentionally discourage the use of active strategies and promote cognitive offloading instead [17]. This underscores the need to design AI-supported learning tools that do not replace but rather support learners' self-regulatory processes during learning activities, such as when students interact with commercial LLMs to solve a code or perform a writing task.

2.3 AI Tutoring and SRL Support: Current Tools and Research Gaps

This section provides an overview of existing AI-mediated learning tools that support student learning through LLMs, including chatbots and tutoring assistants. The focus is on examining their functionalities, target audiences, and pedagogical focus to identify research gaps, specifically regarding their relevance for higher education and their support for SRL.

Over the last years, LLMs have rapidly emerged and been increasingly adopted by students to support their learning activities, even though most of these systems were not originally designed for educational purposes. They vary in purpose, audience, and pedagogical intention. The following section presents a selection of popular tools and discusses their applicability to higher education and SRL.

ChatGPT (OpenAI) ChatGPT is a general-purpose conversational AI that can write text, answer questions, and assist with various academic and everyday tasks [38]. Despite being commonly used by university students to draft written content or clarify complex concepts [1, 39], it is not designed to support students in the use of their SRL skills. It does not possess features to help students set learning intentions, monitor progress, or reflect on their learning style. Recently OpenAI has released "Study Mode," a feature intended to support students by offering structured explanations, quizzes, and step-by-step guidance [21]. While this represents a shift toward more learning-oriented interaction patterns, Study Mode does not explicitly support SRL strategies. Instead, it aims to assist with content understanding but does not help students plan their learning, monitor their strategies, or reflect on their prompting behaviour. Moreover, adoption among university students remains limited, and those who use it tend to already be highly self-regulated learners who actively seek structured support [22, 23]. This suggests that optional,

mode-based learning features may primarily attract students who are already motivated to regulate their learning, raising broader questions about how SRL support can reach students who do not actively seek it. Therefore, the limitations identified in existing LLMs with respect to SRL scaffolding still remain, and there is still a need for tools like ChatWise that offer real-time metacognitive feedback aligned with SRL theory.

Khanmigo (Khan Academy) Khanmigo is an AI-powered tutor developed by Khan Academy that can interact with students in a more educationally enriching way. Khanmigo guides students through problems, prompts reflection, and nudges step-by-step explanation [19]. However, Khanmigo is specifically designed for use among K-12 students (children and youth typically in the range of ages 5-18) and its pedagogical composition is not necessarily accommodated to the learning preferences of university students who often study on their own.

Elicit Elicit is an assistant software for research students aimed at helping users automate a portion of the academic research process, such as question refinement and literature review [40]. It is more academically oriented than ChatGPT but on a restricted, task-specific basis. It has some features of support for higher education workflows but lacks wider support for student self-regulation or metacognitive development.

GrammarlyGO and AI writing assistants Tools like GrammarlyGO give instant comments on tone, grammar, and clarity [20]. They are helpful for writing but are not learning-oriented and do not promote greater knowledge or learning strategies. They improve output quality but do not engage with the process of learning or self-regulation.

Embedded AI in productivity software (e.g., Notion AI, Google Docs AI) These productivity-focused applications also increasingly offer AI assistance for summarizing, writing, or brainstorming [41, 42]. Convenient as they are, these features are typically limited to surface-level benefits and do not offer structured pedagogical support. Their use is student-driven and lacks any built-in mechanisms for encouraging reflection or planning.

Despite the rapid development of generative AI technology in learning, there is a critical shortfall in the availability of systems that focus on encouraging SRL with generative AI technologies, particularly within the university learning environment. Most available solutions are for a general public or for younger learners and fail to tackle the unique issues university students face, such as integrating challenging information, critical thinking, and long-term learning goals.

In addition, the above tools are primarily task-oriented and reactive. They respond to the input provided by the user but fail to support the user in processes such as goal-setting, choosing strategies, or reflection. This absence of pedagogical structure limits their capacity to foster independent learning skills, needed to attain academic success at university level [12, 15].

In summary, these findings highlight the need for adaptive technologies that integrate the use of generative AI with purposeful design principles based on SRL theory. Such support could help students not only complete academic tasks but also learn from the learning process itself, developing the habits of mind that characterize effective, self-directed and autonomous learners.

2.4 Research Gaps

Although LLMs like ChatGPT are now deeply integrated into STEM students' study practices [8, 1], their widespread use raises concerns about students' engagement in SRL and critical reflection. Research increasingly reports risks of cognitive offloading and "metacognitive laziness"—where learners delegate key cognitive processes to AI [11]. These tendencies undermine SRL, a core driver of academic success in STEM and programming education [12, 35, 32].

While some studies have explored how students use LLMs to complete tasks, less is known about how they regulate their learning *in real-time* during LLM interactions. Existing tools like ChatGPT or Claude are designed to deliver answers, not to scaffold students' planning, monitoring, or reflection processes [17, 6]. Likewise, pedagogically oriented systems such as Khanmigo or GrammarlyGO tend to support only narrow tasks or younger learners, with limited attention to university-level SRL strategies such as effort regulation or elaboration.

This highlights a deeper pedagogical and design challenge: commercial LLMs currently encourage efficiency, not metacognitive engagement. Yet for students to benefit meaningfully from these tools, their interactions must support—not substitute—self-regulatory thought processes. There is a critical need for systems that do not merely deliver information but help students frame learning goals, evaluate outputs, and refine strategies during AI use.

These issues point to a specific research gap: despite the widespread use of generative AI in higher education, we lack empirical and design-oriented studies that investigate how to support university STEM students' SRL processes in everyday interactions with LLMs. In particular, no existing work has implemented or evaluated real-time, SRL-aligned metacognitive feedback that enables students to use LLMs more intentionally, critically, and

reflectively. Bridging this gap requires designing support that encourages SRL during students' typical interactions with LLMs, while preserving autonomy and workflow.

Chapter 3

Method

This chapter presents the research design and methodological approach adopted in this thesis, which follows the principles of Design-Based Research (DBR), an iterative methodology combining the development of an educational intervention with systematic analysis in authentic contexts.

Section 3 introduces the overall DBR approach, outlines the five iterative steps guiding the study, and explains why this methodology was selected. Sections 3.1 and 3.2 correspond to the first two DBR steps by describing the identification of practical and theoretical problems and the design of the ChatWise intervention. Section 3.3 details the data sources and data collection process associated with the implementation of the prototype in authentic learning contexts. Sections 3.4 and 3.5 address the later DBR steps by describing the study procedure and the quantitative and qualitative analysis methods used to evaluate changes in students' SRL-related prompting behaviour. Finally, Section 3.6 introduces the framework used to evaluate the impact of ChatWise on students' SRL strategies.

Introducing ChatWise. ChatWise is a browser extension developed in this project to provide real-time, adaptive feedback on students' prompts when they use ChatGPT for study-related tasks. The tool classifies each prompt according SRL strategies such as planning, elaboration, and reflection, and offers visual feedback designed to foster more reflective and strategic engagement with LLMs. Through the combination of technical implementation and pedagogical intent, ChatWise was conceptualized as a research-based intervention that both functions as an educational tool and serves as a data collection instrument, bridging the gap between LLM-assisted learning and SRL theory.

Design-Based Research Approach

This study adopts a DBR approach to design, implement, and evaluate ChatWise. DBR was chosen because it supports iterative design in authentic educational settings and combines practical improvement of the intervention with generation of theoretical insights [24, 43].

Unlike Design Science Research, which primarily focuses on artifact performance and evaluation [44], DBR explicitly targets educational practice and theoretical refinement simultaneously. Because ChatWise is both a technical artifact and a pedagogical intervention, DBR offers the most suitable methodological alignment: it enables iteration within authentic learning contexts and supports the development of design principles grounded in educational theory [25, 24].

Following the DBR principles [25, 24], the project was structured as an iterative, cyclic process consisting of the following five steps:

Step 1 Identifying practical and theoretical problems,

Step 2 Designing and developing a solution (ChatWise),

Step 3 Implementing and refining the prototype through iterative testing,

Step 4 Collecting data and analysing user interaction patterns, and

Step 5 Evaluating the impact of ChatWise with a focus on changes in students' SRL strategies, as measured through pre- and post-surveys and comparative analysis of SRL-based prompt classifications before and after ChatWise use.

The DBR approach enabled continuous feedback from students to refine ChatWise while simultaneously collecting data to understand its impact on students' prompting behaviour and SRL engagement when using ChatGPT to support their self-study. The integration of qualitative insights (from open-ended survey responses, user interactions, and in-session behavioural observations — such as how participants reacted to visual feedback or revised their prompts) and quantitative data (prompt classifications and frequency analysis) aims to ensure a comprehensive understanding of both the tool's functionality and its pedagogical implications on fostering students' more reflective and critical engagement with ChatGPT.

Grounding the intervention in real usage contexts within STEM higher education allowed each design iteration to directly respond to students' needs and learning behaviours, strengthening both the practical relevance of the tool

and the theoretical contribution regarding AI-supported learning in higher education.

Research Questions Links to DBR

This thesis explores how adaptive support can enhance SRL in STEM education when students use generative AI tools such as ChatGPT. Guided by the DBR framework, the study is structured around three research questions:

1. **RQ1 (Baseline):** How do university students currently use ChatGPT to support their studies, and to what extent do these uses reflect SRL strategies?

2. **RQ2 (Design):** How can adaptive support be designed within ChatGPT-based tools to scaffold students' planning, monitoring, and reflection processes, thereby strengthening SRL in STEM higher education?

3. **RQ3 (Impact):** How does the use of ChatWise influence students' SRL prompting behaivour during ChatGPT-assisted study tasks?

RQ1 examines the **baseline context**, focusing on how students currently use ChatGPT in their studies and the extent to which these practices reflect SRL strategies. RQ2 addresses the **design-oriented dimension** of the study, exploring how adaptive support can be embedded into ChatGPT-based tools to promote students' planning, monitoring, and reflection processes. RQ3 focuses on the **evaluative dimension**, investigating whether and how ChatWise influences students' prompting behaviour, motivation, and engagement with reflective learning strategies during AI-assisted study tasks.

Together, these questions capture the three core aims of the DBR approach: understanding current practices, designing an adaptive intervention, and evaluating ChatWise's impact on students' SRL with ChatGPT.

3.1 Development: Identifying Practical and Theoretical Problems

The first step of the DBR process focused on identifying both the practical and theoretical gaps that motivated the development of ChatWise. The goal was to understand how existing uses of popular LLMs in higher education currently fail to support SRL among students, and how these insights could inform the design of an intervention.

Identifying the Problem Space

As outlined in Chapter 2, prior research has shown that while LLMs can effectively support problem solving and information retrieval, they often promote surface-level engagement rather than reflection or strategic learning [6, 45]. Preliminary discussions with university teachers and thesis supervisors further indicated that students tend to use ChatGPT primarily to obtain direct answers rather than engaging in deeper reasoning or elaboration.

From a theoretical perspective, the tendency of students to use LLMs primarily for obtaining direct answers rather than for reflection or elaboration indicates a misalignment between current LLM use and SRL theory. According to Pintrich's SRL framework [46], effective learners plan, monitor, and reflect on their learning using strategies such as elaboration, critical thinking, and effort regulation. The lack of adaptive support explicitly designed to foster these behaviours represented both a practical and a theoretical gap that this research aimed to address.

Design Principles Derived from Theory

The pedagogical foundations of ChatWise were guided by SRL principles emphasizing learner autonomy, reflection, and non-intrusive feedback. In this study, these principles were translated into three core design aims:

1. Provide **immediate, formative feedback** that prompts reflection without interrupting the user's workflow.

2. Maintain **learner autonomy and control**, ensuring users can choose whether and how to act on feedback.

3. Foster **intrinsic motivation** through supportive rather than corrective design elements.

To more concretely connect the SRL principles to the system's design, three core design aims were derived from SRL theory: reflection, learner autonomy, and effort regulation. Each of these principles informed a specific implementation choice. The emphasis on **reflection** led to the use of real-time visual cues such as colour-coded underlines and popup explanations that help students monitor the quality of their prompts. The focus on **learner autonomy** motivated the decision to make feedback optional and non-blocking, allowing users to ignore or engage with it as they see fit. The SRL principle of **effort regulation** shaped the inclusion of supportive,

motivational cues (e.g., positive reinforcement for well-formulated prompts) designed to nudge students toward more strategic prompting without creating pressure or friction. Together, these mappings ensured that SRL theory was operationalised through concrete, user-facing interaction patterns.

These principles informed the subsequent technical implementation (see Chapter 4), where they were operationalised through visual feedback mechanisms (Section 4.3.2), SRL-based prompt classification (Section 4.3.4), and iterative design refinement (Section 4.3.3). Based on these principles, the design goals for ChatWise were to promote increased use of elaboration, organisation, critical thinking, and effort regulation, while simultaneously discouraging passive help-seeking and repetitive rehearsal prompts. These goals were central to the pedagogical intention of the tool: the primary objective was to reduce direct answer-seeking (Help-seeking) and surface-level repetition (Rehearsal), while the most prioritised increases were in Elaboration and Effort Regulation, as these strategies were considered most closely aligned with deeper learning and independent problem-solving. Together, these theoretically grounded design principles formed the foundation for the development of ChatWise, guiding both its pedagogical intent and technical structure in the following stages of the DBR process.

3.2 User Study Design: Designing the Research Intervention

Building on the DBR framework introduced in Section 3, this chapter describes how the research intervention was designed, implemented, and evaluated across multiple stages of the DBR cycle. The study was conducted in two iterative phases: (1) the design and development of the ChatWise prototype, and (2) an exploratory user study evaluating its impact on students' SRL-related prompting behaviour when interacting with ChatGPT. Together, Sections 3.2–3.6 operationalise the DBR cycle by linking problem identification, intervention design, data collection, and evaluation. The study was conducted in May–June 2025.

The primary aim was to examine not only how students interacted with ChatWise, but also whether this adaptive metacognitive support encouraged more reflective and critical prompting behaviour aligned with SRL strategies such as elaboration, effort regulation, and critical thinking, as operationalised in the system's prompt classification scheme (see Section 4.3.4 and [6]).

3.2.1 Participants

A total of 17 university students from different STEM programs at KTH participated in the study. All participants had prior experience using ChatGPT or other LLM-based tools for academic purposes. A convenience sampling approach was adopted to ensure participants represented authentic users of LLMs in higher education contexts. Each participant was compensated with a small gift voucher for their participation.

3.2.2 Study Overview

Participants were asked to use ChatWise while completing study-related tasks within the ChatGPT interface. This design allowed observation of *authentic* SRL-related prompting behaviour rather than contrived or task-specific input.

Figure 3.1 summarises the study timeline. Pilot checks were conducted during the baseline period to verify functionality prior to releasing ChatWise to all participants.

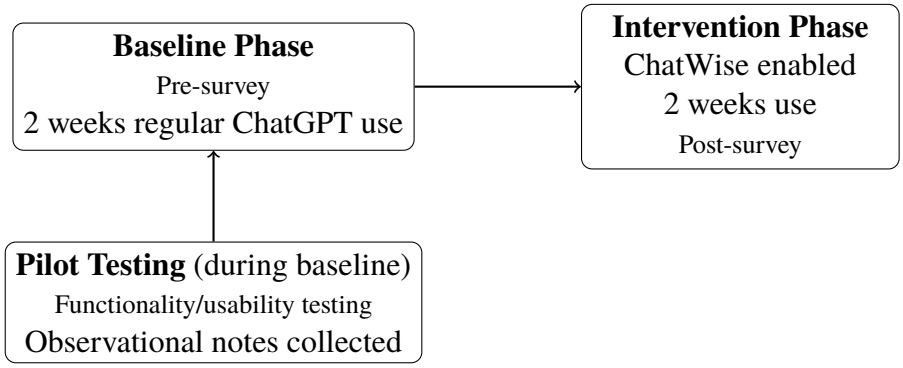


Figure 3.1: Overview of the study design, showing the baseline phase without ChatWise, the intervention phase with ChatWise enabled, and the timing of pre- and post-surveys; pilot study were conducted during the baseline period to verify functionality prior to deployment. The figure situates when different types of data were collected during the project.

During the **baseline phase**, data from students' regular ChatGPT usage were collected prior to introducing ChatWise. This phase aimed to capture how students naturally formulated prompts without feedback interventions. Web scraping was conducted using Selenium and BeautifulSoup to extract anonymized prompt data, which were then classified and exported to CSV format for analysis.

A subset of this pre-ChatWise data was later used to refine the system prompt for post-study analysis (see Section 4.3.4). The same extraction and processing pipeline ensured consistency between pre- and post-intervention datasets, supporting reliable comparison of student behaviour across study phases.

The intervention phase was designed to maintain a balance between ecological validity—allowing students to use ChatWise in authentic study contexts—and experimental control to ensure consistent data collection across participants.

The study focused on three main objectives:

- To observe whether ChatWise influenced the quality and structure of students' prompts,
- To assess students' behavioural responses to visual and textual feedback,
- To explore students' perceptions of usefulness and motivational impact.

3.2.3 Pilot Testing

Before the main evaluation study, a pilot test was conducted with a small subset of participants to identify usability issues and possibly refine design details. Observations from this session informed several adjustments, such as extending popup display time, refining colour contrast, and clarifying feedback messages. Behavioural indicators, including hesitation after red underlines and motivation to achieve "green" feedback, were noted and used to guide final prototype adjustments.

These pilot observations were documented by the researcher to identify usability and interaction issues and were used solely for iterative refinement of the prototype, not included in the empirical data analysis.

3.2.4 Study Variables

To align with the research objectives, the study captured both behavioural and perceptual variables related to SRL engagement:

- **Behavioural indicators:** distribution and change in SRL-based prompt classifications before and after using ChatWise.
- **Perceptual indicators:** perceived usefulness, motivation, and awareness of learning strategies, collected through surveys and optional interviews.

3.2.5 Ethical Approval and Participant Rights

All participants were informed about the purpose of the study, data handling procedures, and their right to withdraw at any time without consequence. Informed written consent was collected before participation. Logged data were anonymized and stored securely, and no personally identifiable information was retained. Ethical pre-approval was obtained in accordance with institutional research ethics guidelines.

3.3 Gathering Empirical Data from the Prototype

The third step of the DBR process focused on the empirical data collection from the functioning ChatWise prototype. The aim of this stage was to gather both system-level interaction data and user-reported experiences to evaluate how students engaged with the feedback mechanisms and whether their prompting behaviour aligned more closely with SRL strategies.

Multiple data sources were combined to ensure triangulation of behavioural, perceptual, and technical evidence.

3.3.1 Data Sources

The following data sources were used in the study:

- **Prompt Logs.** Every prompt submitted during the study was automatically logged, including timestamps, SRL classification (positive, suboptimal, negative, or neutral; see Section 4.3.4 for details).
- **Observational Notes.** During the baseline phase, a small subset of participants took part in researcher-observed pilot interactions with ChatWise to identify potential usability or functionality issues prior to the full intervention. During these observations, the researcher documented visible reactions and interaction patterns (e.g., hesitation after red underlines, satisfaction with green popups, or avoidance of feedback).
- **Surveys.** A pre-survey collected background data on participants' study habits and prior use of LLMs, while a post-survey measured perceived usefulness, motivation, and self-reported changes in prompting strategies.

These data sources provided a comprehensive view of both the quantitative aspects of user interaction and the qualitative dimensions of learning perception.

Survey design. The pre- and post-surveys were designed to capture both behavioural and perceptual aspects of students' interaction with ChatWise. The pre-survey aimed to establish a baseline understanding of participants' study habits, motivation, and prior use of ChatGPT, providing a reference point for evaluating later changes in SRL strategies. It consisted of 10 five-point Likert-scale items and short open-ended follow-up questions addressing these aspects. The post-survey included 12 five-point Likert-scale items together with open-ended questions addressing perceived usefulness, motivation, reflection, and behavioural change after using ChatWise. All Likert items used a 1–5 agreement scale (1 = strongly disagree, 5 = strongly agree), and negatively worded items were reverse-coded prior to analysis. Example items included statements such as "I reflect carefully on how I formulate my questions to ChatGPT" and " ChatWise made me think more carefully about how I use ChatGPT." The items were adapted from validated instruments on SRL and perceived usefulness of AI tools in education [13, 6].

Each survey item was mapped to SRL-related dimensions based on Pintrich's framework [13], such as reflection, planning, elaboration, effort regulation, and motivation. Both the pre- and post-surveys included items on perceived usefulness, while the post-survey additionally included user experience items adapted from Viberg et al. [6].

Full versions of the pre- and post-surveys can be found in Appendix A and Appendix B.

3.3.2 Sampling Strategy

The study focused on students who have been already using ChatGPT for study-related purposes, ensuring that data collection captured authentic user behaviour in realistic educational contexts. This approach aligned with the DBR principle of studying interventions within natural learning environments rather than controlled laboratory settings.

3.3.3 Data Handling and Anonymisation

Data were anonymised and handled according to the ethical procedures described in Section 3.2.5. This ensured full compliance with institutional research ethics standards and data integrity throughout the empirical phase.

The collected dataset provided the foundation for subsequent analysis, described in the next section.

3.4 Collecting and Analysing User Interaction Data

The fourth step of the DBR cycle focused on the practical enactment of the user study, implementing the data collection plan described in Section 3.3. This section outlines the concrete procedure followed during the study, including timing, participant actions, and data recording steps.

3.4.1 Test Environment

The study was conducted remotely, with participants using their personal computers in their naturalistic study environment to ensure authentic behaviour. Each participant used the ChatWise browser extension integrated directly into the ChatGPT interface (see Section 4.1 for a detailed description of the tool). This setup allowed participants to work naturally while the system logged all interaction data in the background.

3.4.2 Procedure Steps

The study followed a fixed sequence to ensure reproducibility and alignment with the DBR methodology:

1. **Information and consent.** Participants were first informed about the study and confirmed their consent before beginning the session (see Section 3.2.5).

2. **Pre-survey.** Participants completed a brief questionnaire capturing baseline data on their study habits, prior LLM use, and SRL strategies.

3. **Baseline task phase (without ChatWise).** Before installing the extension, participants used ChatGPT for their regular study-related tasks over a two-week period. All prompts from this phase were logged and later classified according to SRL categories to establish a behavioural baseline for comparison with the subsequent intervention phase.

4. **Setup.** Participants were asked to install the ChatWise browser extension (or verified that it was active), logged in, and confirmed that prompt logging was enabled.

5. **Intervention task phase (with ChatWise).** Participants worked on their own study-related tasks using ChatGPT (e.g., clarifying theoretical concepts, writing explanations, debugging code) for 2 weeks. They interacted with ChatWise as follows:

    - Typed prompts as usual; ChatWise classified each prompt in real time.
    - Observed visual cues such as coloured underlines and popup messages providing immediate feedback.
    - Were free to revise or rephrase prompts after receiving feedback, though revisions were entirely voluntary.

6. **End of interaction.** Each interaction ended when participants completed their task or reached a natural stopping point.

7. **Post-survey.** After completing the two-week intervention, participants filled in a single post-survey measuring perceived usefulness, motivation, and perceived learning with ChatWise. They were also invited to provide open-ended reflections on their experience.

All prompts and classification results were logged automatically during the task phase. Survey responses complemented this system-generated data by capturing students' self-reported perceptions. In addition, observational notes from a small number of researcher-observed pilot interactions conducted prior to the intervention provided qualitative insights into early usability and interaction patterns.

3.4.3 Data Collected During the User Study

During the study, two main types of data were collected:

- **System data:** all submitted prompts.
- **Survey data:** pre- and post-surveys with Likert-scale items and open-ended responses.

For clarity, negatively phrased Likert items were reverse-coded using $x' = 6 - x$ (1–5 scale) before any composite analyses; descriptive tables in Results show raw item means.

These data sources formed the empirical basis for the quantitative and qualitative analyses presented in the next section (Step 5), where SRL results are reported as phase-specific percentages (within pre and within post) per participant, as defined in Section 3.5.1.

3.5 Evaluating the Impact on Students' SRL Strategies

The fifth step of the DBR cycle focused on analysing and interpreting the data collected during the user study. The purpose of this phase was to evaluate the extent to which ChatWise influenced students' SRL prompting behaviour when interacting with LLMs.

This evaluation combined behavioural data from prompt logs with self-reported survey data, using descriptive comparisons of pre- and post-survey Likert-scale responses to examine perceived changes in students' use of SRL strategies.

Quantitative data (prompt logs) were analysed using descriptive statistics to examine changes in SRL category distributions before and during the ChatWise condition. The analysis compared relative frequencies of each SRL category within and across participants.

Qualitative data (open-ended survey responses) were analysed through reflexive thematic analysis [47], which involved iterative coding and theme refinement to identify patterns in participants' perceptions and reflective learning processes. The combination of these methods enabled a mixed-method triangulation that captured both behavioural and perceptual shifts in SRL strategies.

3.5.1 Quantitative Analysis

The quantitative analysis focused on measurable indicators of change in students' prompting behaviour. Specifically, we analysed the following metrics:

- The number of prompts classified into each SRL category

- Changes in prompt types over time (pre- and post-intervention comparison)

Operationalisation. For each participant i, strategy s, and phase $P \in \{\text{pre}, \text{post}\}$, we computed the phase-specific proportion:

$$p_{i,s,P} = 100 \times \frac{n_{i,s,P}}{N_{i,P}},$$

where $n_{i,s,P}$ is the number of prompts coded as s in phase P, and $N_{i,P}$ is the total number of prompts for participant i in phase P. The heatmap (see Fig. 5.1) visualises percentage-point differences $\Delta_{i,s} = p_{i,s,\text{post}} - p_{i,s,\text{pre}}$.

The analysis focused on comparing relative frequencies of SRL categories before and after the intervention to identify shifts in prompting behaviour. Patterns of change were interpreted descriptively to explore potential improvements in students' reflective engagement. In line with the design goals of ChatWise, the analysis specifically examines whether elaboration, organisation, critical thinking, and effort regulation increase during the intervention phase, and whether passive help-seeking and shallow rehearsal decrease.

Table 3.1 summarises the SRL categories used in this study, including their operational definitions, typical prompt examples adapted from Viberg et al. (2025) [6], and prototypic ChatWise feedback used during classification. These categories form the basis for the quantitative analysis presented in Chapter 5.

3.5.2 Qualitative Analysis

The qualitative analysis explored participants' perceptions beyond what was measurable in the prompt logs. Insights were drawn from open-ended survey responses, which were coded and analysed thematically to identify recurring patterns related to reflection, motivation, and perceived usefulness.

Themes were developed inductively to capture:

- How students engaged with the feedback mechanisms (e.g., reactions to colour cues or popups),
- Perceived usefulness of ChatWise as a study support tool,
- Reported or observed behavioural shifts, such as greater attention to prompt formulation or increased effort to achieve positive feedback.

Table 3.1: SRL strategies, descriptions, and student prompt examples adapted from [6], with added prototypic ChatWise feedback.

SRL Strategy	Description	Example Prompt	Prototypic ChatWise Feedback
Rehearsal (RE)	Students repeat prompts or re-ask identical questions without adding new information.	n.a.	*"This prompt repeats earlier content without adding new details. Try specifying what part you still find unclear to deepen your understanding."*
Organisation (OG)	Students restructure, rephrase, or summarise information.	Rewrite this passage to improve sentence structure. Also, add definitions of terms wherever needed.	*"Great job improving the text's clarity and structure — this demonstrates strong organisational skills."*
Elaboration (EL)	Students seek to understand learning material by making connections between concepts.	How do we know that there are no RHP poles? How did you calculate that?	*"Great job asking for clarification on a specific part of the solution. That shows you're engaging with the material actively."*
Help-seeking (HS)	Students ask for definitions, ideas, hints, or direct solutions.	What is webassembly?	*"This prompt is too vague and doesn't show any attempt to engage with the problem. Try explaining what you've already tried or what you find difficult."*
Effort regulation (ER)	Students request explanations, clarification, or additional steps to understand the problem or solution.	I have this line: [...] ...And this warning: [...] How can I check if that key has a value assigned in the dictionary?	*"You're actively trying to troubleshoot a problem and seeking a solution based on your observations, which is a great effort regulation strategy."*
Critical thinking (CT)	Students evaluate, question, or doubt information; they challenge assumptions.	If Enc(k, m) = m xor k and Enc^{-1}(k, c) = c xor k^{-1}, is then Enc^{-1}(k, c) = m?	*"Great question - you're trying to understand why a method that works in one context might not work in another. That shows critical thinking."*

These qualitative findings complemented the quantitative results by explaining how and why certain prompting behaviours changed during the study.

3.5.3 Software Tools

Data analysis and visualization were performed using Python, including the generation of figures for quantitative results. Google Sheets was used for data cleaning and basic descriptive summaries. Qualitative responses were coded manually using an inductive thematic approach supported by Google Sheets. Tables and figures were formatted and compiled using LaTeX for inclusion in the final report.

3.6 Evaluation Framework

Building on the analyses presented in the previous section, this framework outlines how the results were interpreted and evaluated in relation to the research questions, ensuring a systematic link between empirical findings and the study's theoretical aims. We evaluate the impact of ChatWise in relation to RQ1 and RQ3 along two complementary dimensions: (1) behavioural indicators, focusing on the distribution of prompt types and

their alignment with SRL strategies, and (2) perceptual indicators, reflecting students' reported usefulness, motivation, and reflective engagement after using the tool. These outcomes are mapped to SRL strategies (EL, ER, CT, HS, OG, RE) and compared to baseline data collected prior to the introduction of ChatWise.

Chapter 4

Implementation

This chapter presents the implementation of ChatWise, the browser extension developed to support SRL during students' interactions with ChatGPT. Building on the design principles and pedagogical foundations described in Chapter 3, it describes how the system was realized in practice — from overall architecture to core features, design iterations, and technical challenges. The chapter concludes with a system demonstration showing how these components work together in practice.

4.1 System Overview

ChatWise has been implemented as a browser extension designed to enhance the educational utility of ChatGPT by integrating real-time feedback on user prompts. Its architecture consists of three primary components: (1) the front-end browser extension, (2) a cloud-hosted backend server, and (3) a document-based database.

The front-end extension injects scripts into the ChatGPT interface, allowing it to monitor and classify user prompts as they are submitted. When a user types a prompt, the extension sends it to the backend server, which in turn calls a language model API to evaluate the prompt's alignment with SRL strategies. Figure 4.1 illustrates the end-to-end process and how the system components interact.

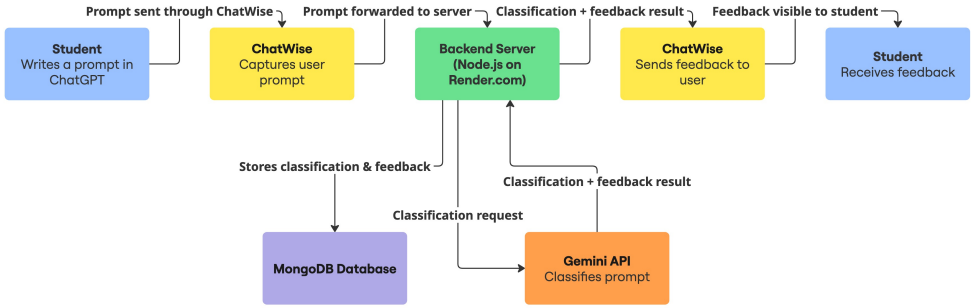


Figure 4.1: System architecture and data flow in ChatWise, illustrating how user prompts are captured in the browser extension, classified via the backend server, and returned as real-time visual feedback to the user.

As shown in Figure 4.1, prompts are processed through the backend server and classification API before the resulting feedback is returned to the user via the ChatWise extension.

The server, hosted on Render.com, processes incoming prompts using a system prompt-based classification method via the Gemini API. The server then returns a classification result, which the extension uses to provide immediate visual feedback in the form of coloured underlines and pop-up suggestions.

All interactions are logged to a MongoDB database, where data is stored for both real-time use (such as enabling user login sessions) and for later analysis. This data includes prompt texts, timestamps, classification results, and user identifiers.

Together, these components form a lightweight, modular system capable of real-time interaction with users while maintaining scalability for future iterations. The architecture is designed to be extensible, allowing for potential integration with additional learning platforms or AI models.

4.2 Core Features

This section describes the main functional components that together form the core functionality of ChatWise. Each feature contributes to the system's ability to capture, classify, and respond to user prompts in real time, providing a foundation for the pedagogical feedback mechanisms introduced later in the chapter.

4.2.1 Prompt Monitoring and Capture

When users interact with ChatGPT, ChatWise captures each submitted prompt via a JavaScript listener in the browser Document Object Model (DOM). The captured text is sent to the backend for classification and feedback, operating in parallel with ChatGPT's response without disrupting the normal workflow.

4.2.2 Prompt Classification Engine

Using the Gemini API and a preconfigured system prompt, ChatWise evaluates each user prompt based on its alignment with SRL strategies. The classification maps prompts into four categories: positive (SRL-aligned), suboptimal (vague or shallow help-seeking), negative (unreflective delegation), or neutral (non-academic or contextless). Classification is performed on the backend to ensure secure API handling and consistent model behaviour across users.

4.2.3 Visual Feedback System

Based on the classification result, the extension provides immediate, non-intrusive feedback via:

- **Underlines:** on specific words or phrases identified by the classifier (e.g., red for negative prompts, green for positive). These highlights correspond to textual elements detected as relevant by the underlying classification system (see Section 4.3.4),
- **Pop-up messages:** providing either constructive suggestions for improvement or positive reinforcement when prompts demonstrate effective SRL strategies.

This visual layer was designed to be supportive rather than corrective, allowing users to reflect without disrupting their workflow.

4.2.4 Gamified Motivation Cues

ChatWise includes lightweight gamified cues designed to make reflective prompting more engaging. When a prompt aligns with SRL strategies, the system displays a green popup with positive feedback, reinforcing effective prompting behaviour. These cues are integrated directly into the feedback system and operate in real time alongside ChatGPT's responses.

4.2.5 User Interaction Logging

All user actions — prompt texts, classification results, and timing data — are logged securely to support retrospective analysis. The logged data also enables later analysis of user behaviour for research and refinement purposes. While the current implementation does not analyse re-prompting frequency, the collected data would enable such analyses in future iterations.

4.2.6 Popup Prompt Coaching

When prompts are classified as *suboptimal* or *negative*, the system provides short, adaptive coaching messages to guide users toward more reflective phrasing. The feedback is concise and supportive, aiming to enhance self-awareness without interrupting the user's workflow.

These core features work in tandem to subtly guide users towards better prompting practices and a deeper understanding of how they interact with generative AI for academic tasks.

4.3 Design Process and Iterative Development

The design and implementation of ChatWise followed an iterative, user-centered process aligned with the principles of DBR. This process aimed to operationalize the pedagogical goals described in Chapter 3 by translating SRL principles into concrete design features, interface behaviours, and feedback mechanisms.

4.3.1 User Interface Design Principles

The design of ChatWise was inspired by existing feedback systems such as *Grammarly* [48], which underline questionable text and provide corrective suggestions on hover (see Figure 4.2). This interaction model combines immediate visual cues with just-in-time feedback — two elements that were adapted to the context of prompt-based interaction with LLMs.

In ChatWise, this concept was implemented through colour-coded underlines that visually signal prompt quality, combined with popups that appear automatically after classification and can later be revisited by hovering over the highlighted text. These visual cues were intentionally subtle, aiming to prompt reflection without disrupting the user's workflow. The design

thereby aligns with SRL principles of autonomy and self-awareness, allowing users to remain in control of their learning process while receiving formative, non-intrusive guidance.

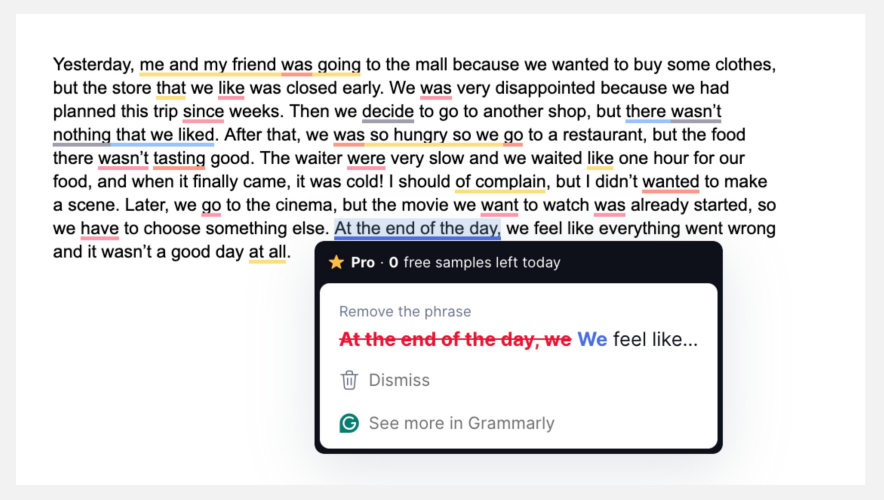


Figure 4.2: Example of how *Grammarly* provides in-text feedback.

4.3.2 Feedback and Motivation Mechanisms

To encourage self-regulation and sustained engagement, ChatWise integrates visual feedback and light gamification. Colour-coded underlines communicate prompt quality at a glance, while supportive popups provide immediate, formative feedback in real time. Prompts reflecting shallow help-seeking behaviour (e.g., "Solve X for me") are underlined in red and accompanied by a red feedback popup, while prompts that demonstrate elaboration or critical thinking (e.g., "Compare two solutions and explain the advantages of each") are underlined in green and trigger a supportive message.

The system distinguishes between four categories of prompts: *positive* (SRL-aligned), *suboptimal*, *negative*, and *neutral*. Each category is associated with distinct feedback text and colour coding, designed to nudge users toward more reflective prompting behaviour.

This design promotes intrinsic motivation and supports learner autonomy by rewarding effective prompting behaviour without enforcing compliance or disrupting workflow.

4.3.3 Iterative Prototyping and Refinement

Development proceeded through several iterative stages informed by user feedback and observations made during the design and testing process. The initial design was developed using Human-Computer Interaction (HCI) principles, particularly Nielsen's usability heuristics [49], emphasizing visibility of system status, consistency, and user control.

The design process began with an initial mockup (Figure 4.3) illustrating the intended feedback flow and user interaction patterns. The purpose of this mockup was to translate the theoretical SRL principles, reflection, autonomy, and minimal disruption, into a tangible interface concept.

The mockup focused on establishing the core mechanics: visual indicators (coloured underlines and small icons such as checkmarks or crosses) representing prompt quality, together with hover-triggered feedback popups offering context-sensitive advice. At this early stage, no automatic popups were included — feedback was accessible only through hovering, as the goal was to test the feasibility of subtle, non-intrusive visual cues.

One initial pilot test with a a small subset of users served to refine usability aspects. Observations from this session influenced design decisions such as extending popup display time, improving feedback clarity, and refining colour contrast. Behavioural indicators, such as hesitation after red underlines or motivation to achieve "green" feedback, were noted and informed subsequent iterations.

The design evolved through three major stages:

- **Initial mockup:** A static interface concept focusing on layout and visual hierarchy.
- **Early prototype:** A partially functional browser extension without login functionality and with inconsistent feedback behaviour.
- **Refined prototype:** A stable and fully integrated system implementing consistent colour coding, popup timing, and feedback text.

Feedback was gathered through informal think-aloud sessions and a pilot test. Pilot testing also informed final design decisions, such as feedback timing, colour contrasts, and popup persistence. An early design idea to delay the AI's response (to encourage reflection) was discarded after pilot testing, as participants found it disruptive. Instead, motivational design elements were prioritized to make reflective behaviour feel rewarding rather than corrective.

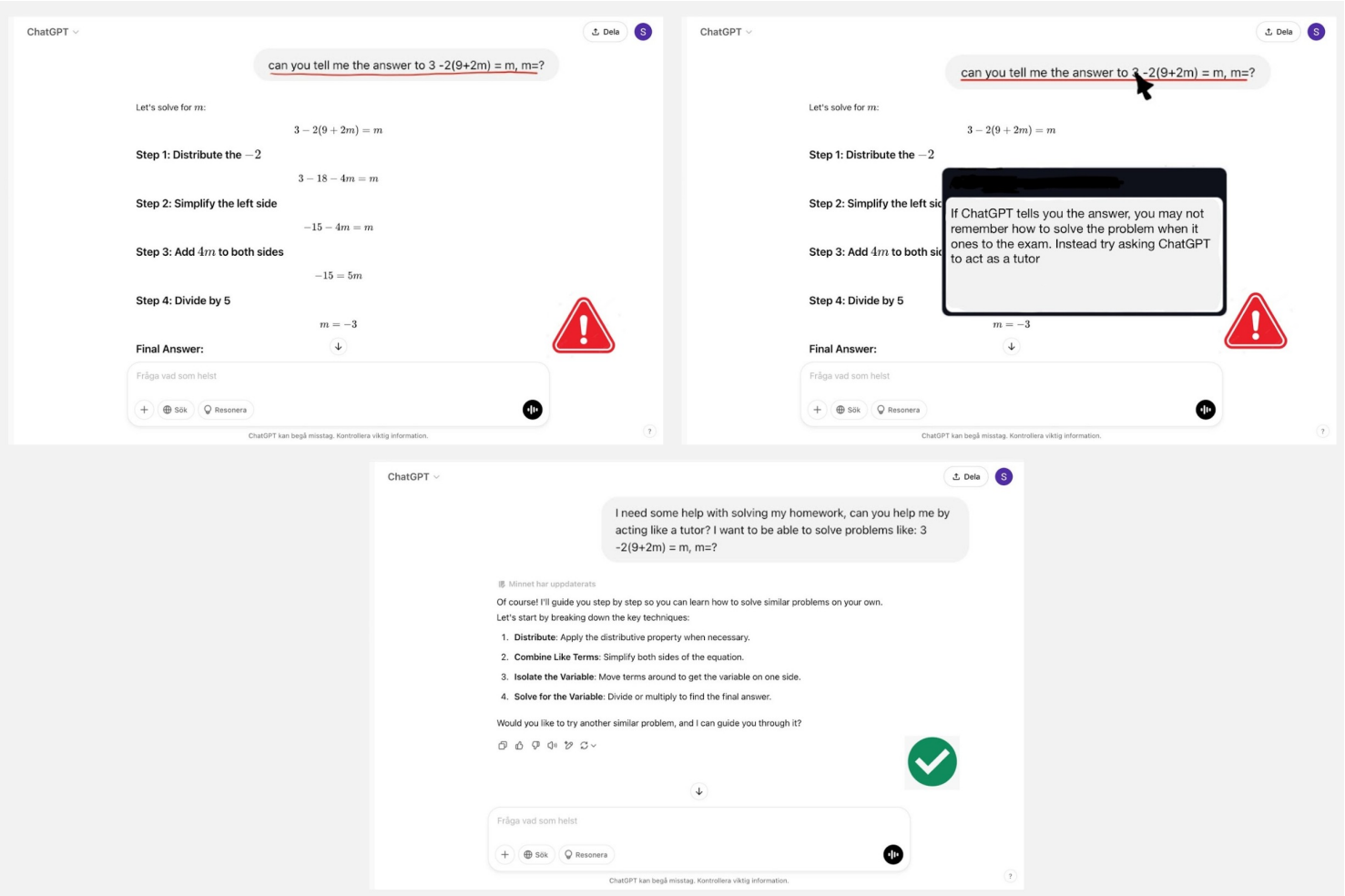


Figure 4.3: Initial mockup design for ChatWise. The concept introduced underlining and hover-based feedback to support reflection without interrupting workflow.

4.3.4 Prompt Classification and Internal Validation

Prompt classification in ChatWise was implemented using the Gemini API. The classification logic was informed by annotated datasets from previous SRL studies [13, 6], where student prompts were manually categorized according to SRL strategies such as elaboration, effort regulation, and critical thinking. A custom system prompt was iteratively refined to ensure that Gemini's classifications aligned with the SRL categories defined in these studies.

System prompt description. The classification logic was implemented

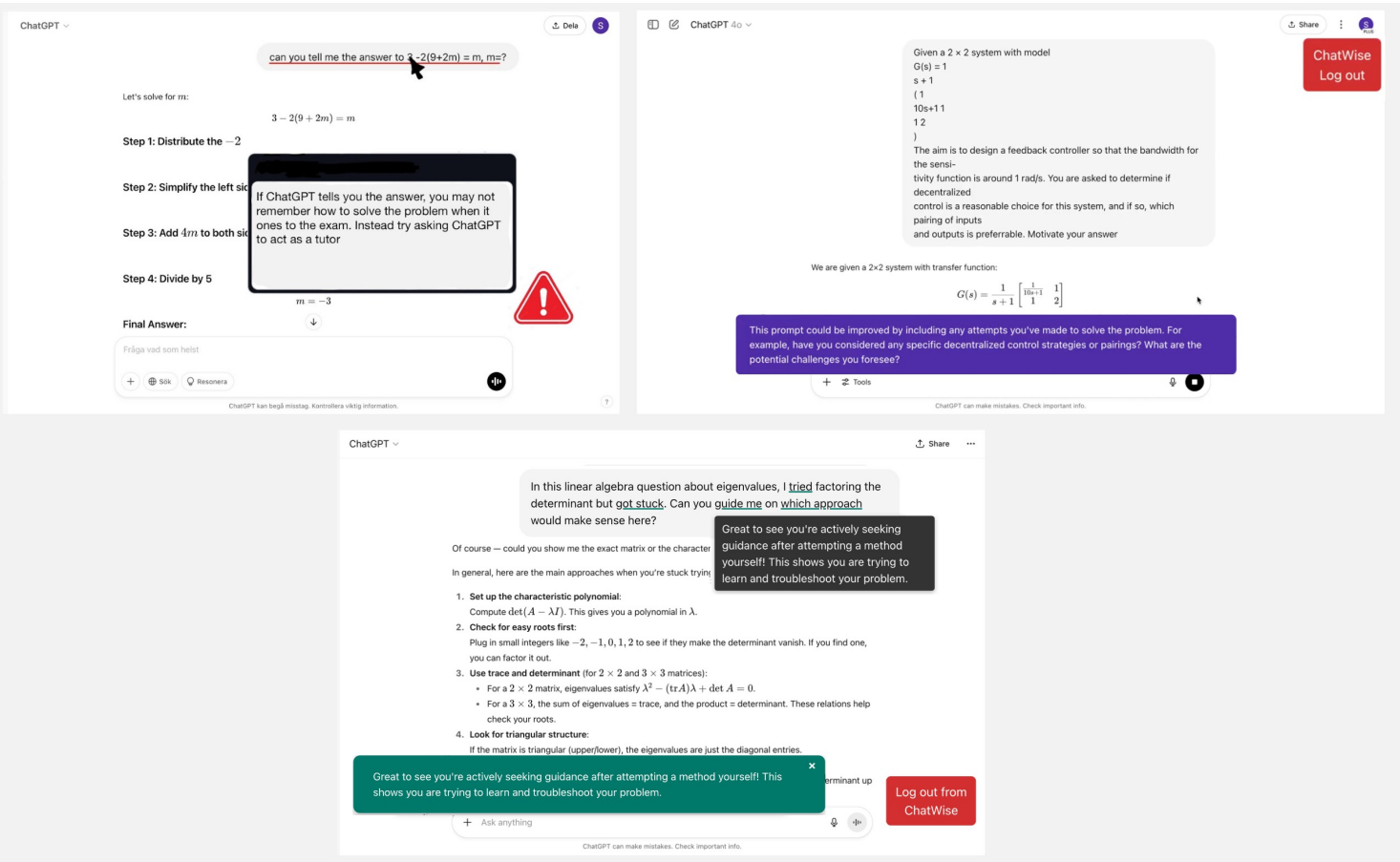


Figure 4.4: Evolution of the ChatWise prototype: (a) initial mockup, (b) early prototype, (c) refined version with improved feedback and gamified cues.

through a fixed system prompt sent to the Gemini API. The system prompt instructed the model to evaluate each user input based on its educational quality and alignment with six SRL strategies: *rehearsal, organisation, elaboration, effort regulation, critical thinking,* and *help-seeking* (See 3.5.1). It specified four possible classification types: `POSITIVE`, `SUBOPTIMAL`, `NEGATIVE`, and `NEUTRAL`, each returned in a JSON format together with explanatory feedback, reasoning, and trigger words from the original prompt. This design allowed ChatWise to provide structured and interpretable responses to users while maintaining consistency in the data logged for later analysis. A full version of the system prompt is included in Appendix C.

An internal validation test was conducted using a manually labeled dataset to assess classification performance. Table 4.1 shows the initial confusion matrix, revealing weaknesses in identifying elaboration, critical thinking, and

rehearsal. These issues were traced to the model's lack of conversational context awareness, as it analysed each prompt in isolation. This initial version of the system prompt was the one integrated into ChatWise during the user study.

Table 4.1: Initial confusion matrix of the system prompt's SRL classification.

True label →	**EL**	**ER**	**CT**	**HS**	**OG**	**RE**
Elaboration	**4**	2	0	14	0	0
Effort regulation	1	**12**	0	7	0	0
Critical thinking	5	5	**3**	6	0	0
Help-seeking	0	2	0	**15**	3	0
Organisation	0	0	1	2	**16**	0
Rehearsal	1	2	1	12	0	**0**

After the study, the system prompt was further refined through iterative prompt adjustments aimed at clarifying distinctions between elaboration and help-seeking, resulting in improved classification accuracy (Table 4.2). This refinement was applied only for post-study analysis of user data and not in the deployed prototype. The process demonstrates how empirical testing informed iterative improvements in line with the DBR methodology.

Table 4.2: Confusion matrix after refinement of the system prompt.

True label →	**EL**	**ER**	**CT**	**HS**	**OG**	**RE**
Elaboration	**7**	0	0	11	0	2
Effort regulation	2	**14**	1	2	0	1
Critical thinking	7	2	**6**	2	0	2
Help-seeking	1	1	1	**13**	2	0
Organisation	3	0	0	1	**14**	1
Rehearsal	4	1	3	4	1	**6**

Technology Stack and Key Decisions

ChatWise is implemented using a modular architecture combining front-end browser scripting with a lightweight backend API server. The system integrates several technologies to enable prompt classification, real-time feedback, and secure data logging:

- **Front-end:** The browser extension is built with JavaScript and integrates directly into the ChatGPT DOM to capture user prompts and display feedback.
- **Backend:** A Node.js server hosted on Render.com handles communication between the extension, database, and Gemini API. Render.com was selected for its simplicity, cost-efficiency, and free tier support.
- **API:** Google's Gemini API manages prompt classification. It was chosen over OpenAI's API due to lower cost, fewer rate-limit issues, and easier integration during development.
- **Database:** MongoDB stores anonymized user data and classification results. A non-relational structure was chosen for flexibility in handling varying prompt metadata.

4.4 Implementation Challenges

During implementation, several technical issues were encountered and addressed:

- **Rate Limiting and Latency:** Gemini API usage occasionally triggered HTTP 429 rate-limit errors, which were mitigated by implementing exponential backoff and throttling logic.
- **DOM Instability:** Minor updates to ChatGPT's interface occasionally broke extension hooks. These were resolved through manual adjustment of DOM selectors and simplified event-handling logic to improve robustness.
- **Cross-Origin Requests:** Secure communication between the extension and backend server required careful CORS configuration to prevent browser security errors.
- **Popup Timing Conflicts:** Visual feedback overlays sometimes overlapped with ChatGPT's native UI elements. This was addressed by dynamically positioning feedback popups relative to the input field using offset calculations.

Despite these challenges, the system remained stable across multiple user sessions in testing. Its modular design also facilitates future updates or integration with alternative AI services or learning platforms.

4.5 System Demo

This section demonstrates the operational flow of ChatWise through a practical walkthrough of how the system functions in use. The purpose is to show how the implemented features described in previous sections work together to support SRL when students interact with ChatGPT.

When a user opens ChatGPT with the ChatWise browser extension installed, the system first displays a login interface, as shown in Figure 4.5. Once authenticated, the extension becomes active in the browser and integrates seamlessly into the ChatGPT interface. No additional configuration is required, ensuring a simple and accessible setup for all users. From this point, every prompt entered by the user is simultaneously intercepted by ChatWise as it is sent to ChatGPT. The prompt is forwarded to the backend for classification, and feedback is returned to the user almost instantly, often before ChatGPT's own response appears.

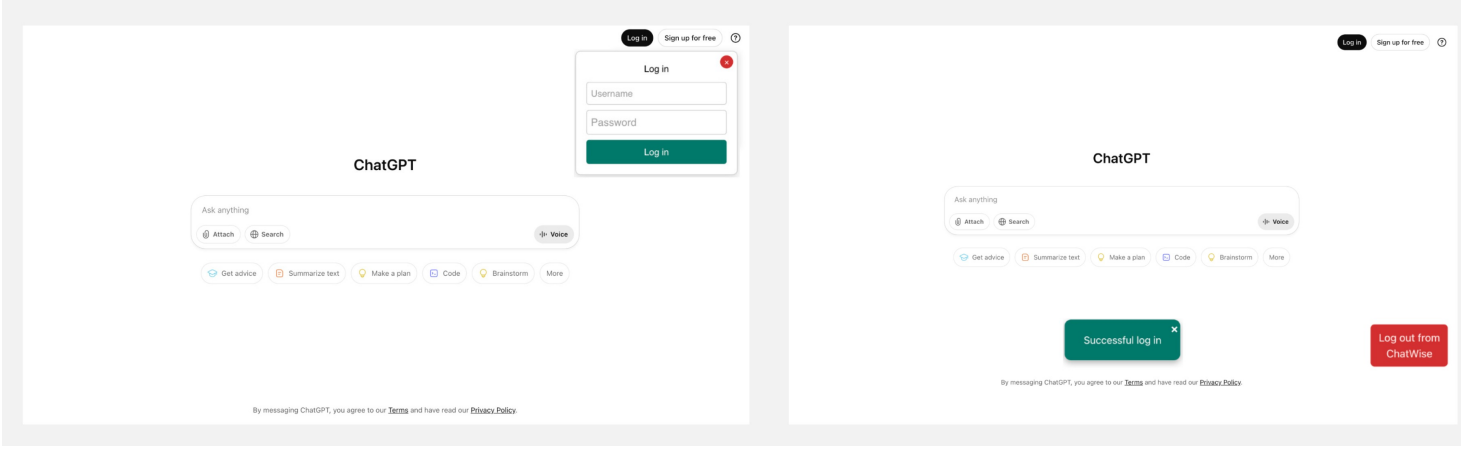


Figure 4.5: Login interface of ChatWise.

After submission, the user receives immediate, non-intrusive feedback directly within the ChatGPT interface (Figure 4.6). Prompts that demonstrate elaboration, effort regulation, critical thinking, or organisation trigger a green popup message with supportive comments, while vague or unreflective prompts result in a red or yellow cue suggesting improvement.

This real-time response helps students become more aware of how they formulate questions and encourages them to rephrase or expand their prompts to promote deeper engagement and reflective thinking. An example of this process is shown in Figure 4.6, where the user receives a contextual popup reinforcing positive prompting behaviour.

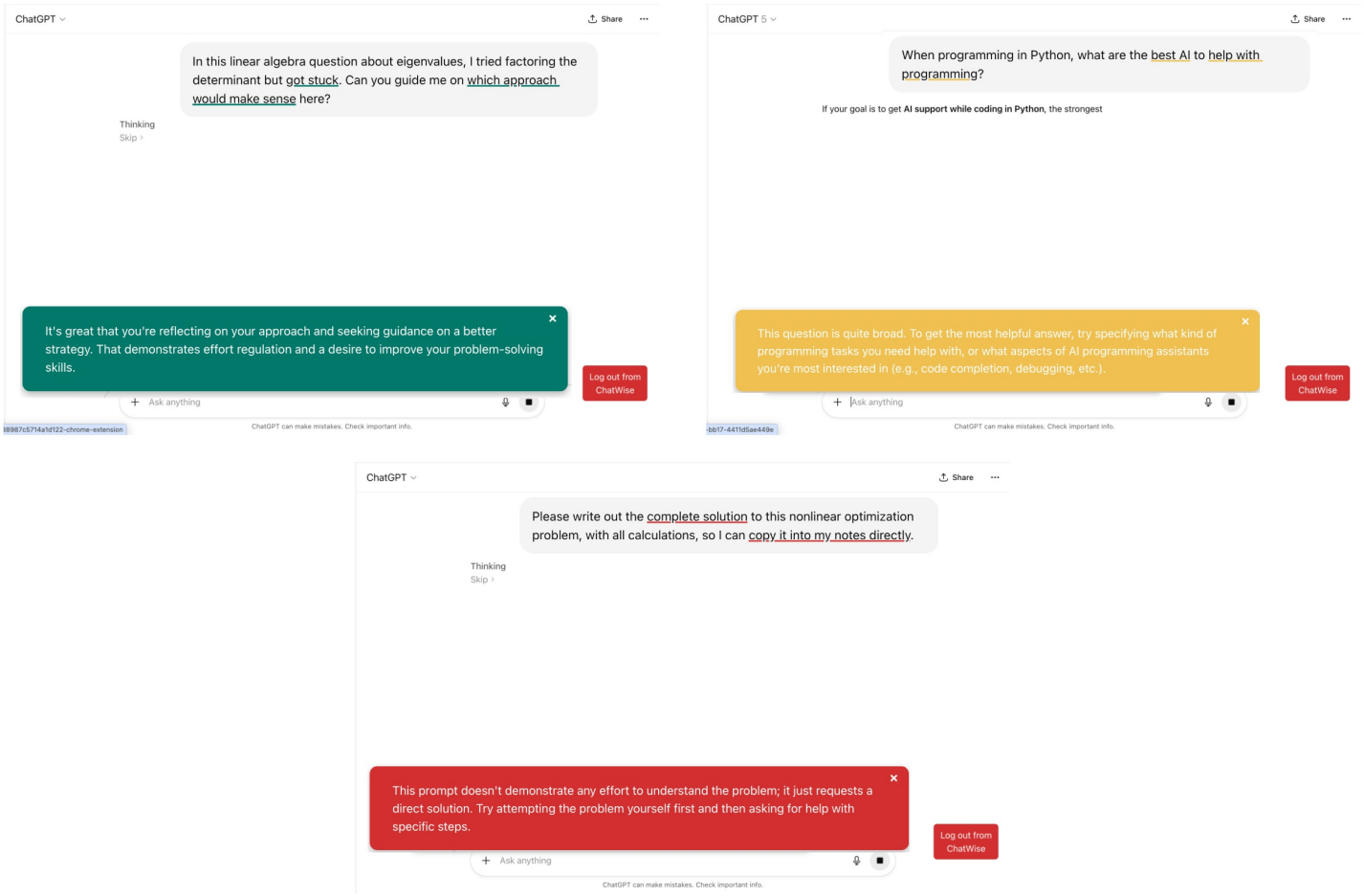


Figure 4.6: Example of popup feedback from ChatWise, shown when a prompt is classified as aligned with an SRL strategy. The feedback is designed to encourage reflection without interrupting the user's workflow.

In addition to popup notifications, ChatWise also supports hover-based feedback, enabling users to explore improvement suggestions without interrupting their workflow (Fig. 4.7). By hovering over the underlined section of their prompt, users can access concise reflective cues, such as "Consider rephrasing this question to include reasoning or comparison." This design fosters autonomy and metacognitive awareness while keeping the learning process fluid and non-disruptive.

Together, these features illustrate how ChatWise translates its pedagogical goals into practical interaction design. By embedding colour-coded underlines, adaptive popups, and hover-based feedback directly within

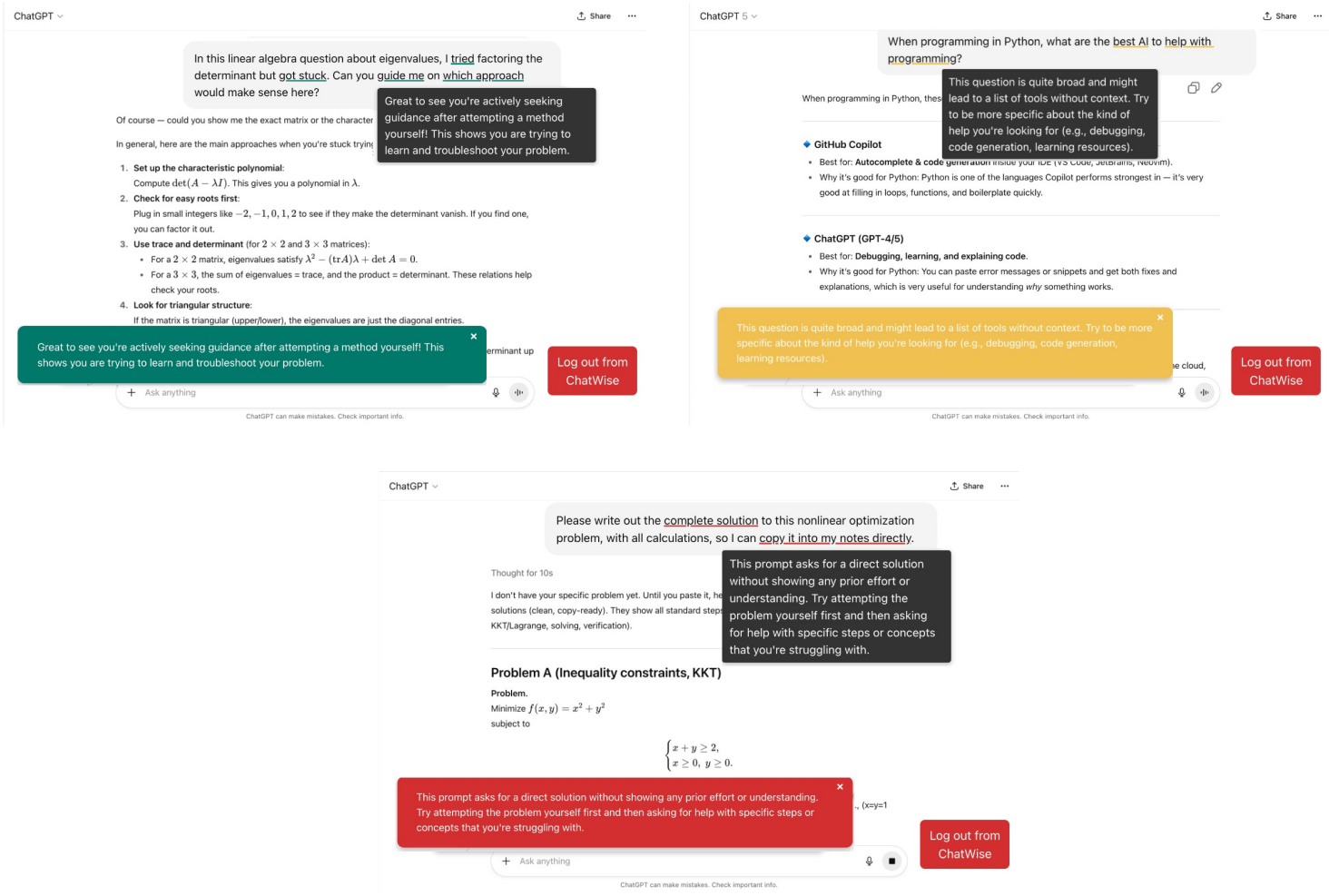


Figure 4.7: Example of hover-based feedback from ChatWise.

ChatGPT's interface, the system provides immediate and supportive guidance aligned with desirable SRL strategies, particularly elaboration, effort regulation, and critical thinking, while discouraging passive help-seeking behaviours.

This seamless integration allows students to reflect and adjust their prompting behaviour in real time, fostering greater self-awareness and autonomy in their interaction with generative AI. The following chapter presents the empirical evaluation of these mechanisms and their observed impact on students' engagement in SRL.

Chapter 5

Results and Analysis

5.1 Participants

In total, 17 students participated in the two surveys conducted in the study (Table 5.1, Table 5.2). Of these, 14 students completed the pre-survey and 10 students completed the post-survey. Nine students took part in both surveys and actively used ChatWise during the overall study period of four weeks. These nine overlapping participants therefore constitute the analytic cohort used for examining prompting behaviour from an SRL perspective.

The participants represented a variety of demographic and academic backgrounds, providing a diverse context for evaluating ChatWise. In terms of gender distribution, the sample was fairly balanced (47 %female, 53 % male). Most of the participants were between 22–27 years old (64 %), though ages ranged from 19 to 32 years, suggesting both early-stage and more experienced university students.

The group was also academically varied. Although 65 % were enrolled at the MSc level and 35 % at the BSc level, the areas of study ranged beyond computer science alone. Approximately one-third studied computer or IT-related programs (CS, ICT, Cybersecurity), but other participants came from media technology, mechanical engineering, production engineering, chemistry, physics, civil engineering, and public health. These distributions refer to all 17 students who participated in the study, not only the analytic cohort of nine.

Although participants differed in academic background and study level, the number of students per program was too small to draw any conclusions about potential differences between educational contexts. As such, the diversity in backgrounds provides descriptive context only, rather than a basis for

Table 5.1: Demographic characteristics of study participants ($n = 17$)

Variable	Category	n	%
Gender	Female	8	47
	Male	9	53
Age	19–21	4	24
	22–24	5	29
	25–27	6	35
	28–32	2	12
Education level	BSc (Bachelor)	6	35
	MSc (Master)	11	65
Field of study	Computer/IT-related (CS, ICT, Cybersecurity)	5	29
	Media Technology / Interactive Media	3	18
	Mechanical / Production / Robotics	4	24
	Chemistry / Physics / Civil Engineering	3	18
	Public Health	2	12

comparative interpretation.

Table 5.2: Number of participants per survey

Survey	n
Survey 1 (Before)	14
Survey 2 (After)	10
Total unique participants	17

Of the 17 unique participants, our analytic cohort comprises the 9 students who both used ChatWise during the study and completed the post-survey. All pre–post analyses are reported for this analytic cohort unless otherwise noted. In total, we analysed $1\,659$ pre-prompts and 839 post-prompts. Results are expressed as percentages within each participant to account for differences in the total number of prompts submitted before and after the intervention.

5.2 Students' Perceived Use of LLMs

RQ link. This section addresses **RQ1 (Baseline)** by describing students' self-reported use of commercial LLMs and associated SRL strategies prior to the

intervention; it establishes the baseline that later sections compare against.

Before the study period with ChatWise, the participants completed a survey assessing their SRL strategies when using ChatGPT. As shown in Table 5.3, students reported relatively high engagement in reflective and deep learning practices, based on their Likert-scale self-assessments. The highest-rated item was "When using ChatGPT, I try to understand why an answer is correct—not just what the answer is" (M = 4.20), followed by "I am motivated to learn deeply rather than just complete assignments" (M = 4.13), and "I reflect carefully on how I formulate my questions to ChatGPT" (M = 3.93).

These high self-reported SRL scores suggest that the participants in this study already viewed themselves as relatively reflective and self-regulated learners when using ChatGPT. This may indicate a degree of self-selection, where students with stronger metacognitive intentions were more likely to volunteer for the study. As discussed in Section 5.3, however, later analyses of their logged prompts reveal that these perceptions do not always align with their actual prompting behaviour.

It is important to note that these are self-reported values, and students may respond in ways they perceive as academically desirable. For this reason, the subsequent section compares these survey responses with students' actual prompting behaviour based on their logged ChatGPT use. This allows for a more accurate assessment of whether their reported strategies align with how they interact with LLMs in practice.

In contrast, items reflecting dependency or disengagement scored lower. For example, students disagreed with "I believe using AI makes me think less when learning" (M = 2.60) and only moderately agreed with "I often depend on ChatGPT to complete tasks or understand concepts" (M = 3.47).

Overall, these findings suggest a balanced use of LLM tools: participants generally preferred autonomous problem-solving before seeking assistance (e.g., "I usually try to solve problems myself before asking for help", M = 3.87).

Table 5.3: Self-assessment of ChatGPT use in learning (scale 1–5)

, ($n = 14$)

Statement	Mean
I mostly ask for direct answers to my problems.	3.53
I reflect carefully on how I formulate my questions to ChatGPT.	3.93
I try to understand why an answer is correct—not just what the answer is.	4.20
I often depend on ChatGPT to complete tasks or understand concepts.	3.47
I actively think about solutions myself before using ChatGPT.	3.60
I usually try to solve problems myself before asking for help.	3.87
I feel confident in my problem-solving abilities without AI assistance.	3.33
Using ChatGPT improves my understanding of the subject.	4.00
I believe using AI makes me think less when learning.	2.60
I am motivated to learn deeply rather than just complete assignments.	4.13

5.2.1 Students' Use of Commercial LLMs

Overall, participants reported high engagement in reflective learning, that is, actively evaluating and making sense of AI-generated answers rather than accepting them at face value (Table 5.3), with the highest means observed for understanding *why* the answer is correct (M=4.20), deep learning motivation (M=4.13), and careful prompt formulation (M=3.93), and lower agreement for items indicating disengagement (e.g., "AI makes me think less", M=2.60).

5.2.2 Thematic Findings

A thematic analysis of the open-ended responses revealed *five* key themes that provide deeper insight into how students approach ChatGPT as part of their learning process.

1. Strategic Use of AI: Balancing Independence and Support Many students described using ChatGPT as a support system rather than a substitute for independent thinking. They often attempt to solve problems on their own before seeking AI assistance. One student explained, *"I usually try to solve a problem, but if I realise I'm stuck I ask ChatGPT to solve it."* (ID9). Another noted that they *"only ask when I don't know"* (ID7), highlighting a pattern of

using ChatGPT as a last resort rather than a first response. Several students framed their use of ChatGPT as similar to asking a TA or a peer, a resource to verify ideas or clarify confusion, rather than a shortcut to answers (e.g.,ID8, ID21).

2. Prompt Reflection and Iterative Inquiry Participants described a clear awareness of the importance of *how* they formulated their questions. Some shared that they began with short prompts and gradually expanded them based on the quality of the initial response, explaining that *"if I get unsatisfying responses, I add more context."* (ID8). Others emphasized a more systematic approach, for example noting that they *"listed all requirements in an organized way and provided pictures to help GPT understand."* (ID4).

Beyond prompt formulation, students described engaging in follow-up questioning to test and explore the model's reasoning. One participant illustrated this metacognitive approach by stating, *"I start with 'solve this' and then ask 'why did you use that variable' or 'what would happen if you changed it.' "* (ID9). This shows a level of reflective practice that goes beyond passive consumption of AI outputs.

3. Deep Learning Orientation Students frequently emphasized the goal of understanding concepts rather than just completing tasks. For example, one participant reflected that *"just knowing the answer is not enough; in the exam I might get a completely different scenario, so I have to understand the theory."* (ID9). Another described how they *"ask ChatGPT for concept explanations and then follow up with specific questions to close knowledge gaps."* (ID8).

This learning-driven orientation was also tied to the recognition that ChatGPT can scaffold understanding when used carefully. As one student put it, *"[... it] helps me build from basic concepts and then test me on the same."* (ID11).

4. Caution and Critical Evaluation of AI Responses A recurring theme was students' critical stance toward the reliability of AI-generated outputs. Many emphasized the importance of cross-checking responses or not fully trusting the output. One student stated plainly, *"I never just copy anything from GPT—I always check if it's true [...]"* (ID7), while others described verifying ChatGPT's answers through textbooks or by reasoning through the problems themselves (e.g., ID9, ID21).

This skepticism often coexisted with appreciation for AI's usefulness. Students viewed ChatGPT as helpful but fallible, useful for generating ideas

or understanding steps, but not an authoritative source. As one participant summarized, *"I depend on it as much as any knowledgeable peer... but I'm equally, if not more, skeptical of the AI's answers."* (ID17).

5. Efficiency, Confidence, and Motivation Students also reported that ChatGPT made their learning process more efficient and manageable, especially when facing new or difficult topics. One participant shared: *"It helps me save time for everything [...] and without it, I would probably fall behind with my studies."* (ID7). Others described how it boosted their confidence, with one noting that *"the emergence of AI lowers my barriers to learning new things and makes me more confident."* (ID4).

However, students also acknowledged the risk of overuse, recognizing that efficiency must be balanced with active thinking. One student cautioned that *"If I always go to ChatGPT and never think about how or why, then I believe I learn less."* (ID9). This reflects an internal tension between using AI as a productivity tool and maintaining personal engagement in the learning process.

Summary

Overall, the pre-survey responses reveal a group of students who approach ChatGPT with both intentionality and reflection. Quantitative items indicate strong motivation toward deep learning and cautious use of AI, while thematic analysis of open-ended responses shows that students view ChatGPT as a tool for supporting—not replacing—their own thinking. Their baseline self-reported SRL behaviours include prompt refinement, critical evaluation, follow-up questioning, and thoughtful decision-making about when and how to seek AI assistance. These initial patterns provide a valuable foundation for interpreting any post-study changes in SRL strategy use.

5.3 Students' Actual Prompting Habits

RQ link. This section addresses **RQ1 (Baseline)** by analysing log-based prompting behaviour prior to ChatWise and contrasting it with self-reports to triangulate the baseline.

While students reported relatively high engagement in reflective and deep learning practices (e.g., trying to understand why an answer is correct, carefully formulating their prompts), the analysis of their interactions with LLMs shows a different pattern. An analysis of prompts collected before

using ChatWise revealed that most interactions with ChatGPT were short, direct, and oriented toward immediate task completion. Table 5.4 shows the distribution of these prompts across SRL strategy categories, indicating that help-seeking and elaboration were the most common, while strategies such as effort regulation and critical thinking were rarely observed.

For a detailed description of the SRL categories, including definitions, example prompts, and prototypic ChatWise feedback, see Table 3.1 in Section 3.5.1.

In practice, students often relied on help-seeking prompts such as asking ChatGPT to "solve" or "explain" problems, with limited evidence of elaboration or critical thinking strategies. For instance, a common pattern was asking ChatGPT to "Write Python code that calculates X" or "Explain concept Y in simple terms." These types of prompts were efficient for obtaining answers but showed little indication of effort regulation, elaboration, or deeper critical reasoning. By contrast, elaboration- or critical thinking-oriented prompts—such as "Compare two possible approaches to solving X and explain which one is more efficient" or "Explain why method A might fail under condition B"—were less prevalent in the logs compared to help-seeking prompts.

Table 5.4: Distribution of students' actual prompting habits before using ChatWise, categorized by SRL strategies. ($n = 9$)

ID	Help-seeking	Rehearsal	Organisation	Elaboration	Effort Regulation	Critical Thinking
ID10	41 %	13 %	4.30 %	34.80 %	4.30 %	2.20 %
ID14	8.50 %	23.90 %	16 %	16.20 %	23.90 %	11.20 %
ID15	11.20 %	12.90 %	46.80 %	19.40 %	4 %	5.60 %
ID17	22 %	14 %	6.00 %	34.70 %	4.50 %	18.60 %
ID19	21.60 %	0 %	11.40 %	38.60 %	3.40 %	35 %
ID20	53.70 %	1.85 %	1.85 %	33.30 %	0 %	9.20 %
ID21	21 %	21 %	26.30 %	26.30 %	0 %	5.30 %
ID22	44.70 %	18.40 %	5.30 %	26.30 %	0 %	5.30 %
ID5	51 %	28 %	1.60 %	18 %	0 %	1.60 %

Note. SRL categories were assigned automatically using our classifier described in Section 4.3.4, which maps prompt content to six SRL strategies (Help-seeking, Rehearsal, Organisation, Elaboration, Effort Regulation, and Critical Thinking).

This discrepancy suggests that students' perceived use of commercial LLMs does not fully align with their actual, observed prompting behaviour. While survey responses reflect an intention to engage critically and thoughtfully, the insights based on the analysis of their logged prompts reveal a tendency toward efficiency-oriented, task-completion behaviour—prioritizing getting answers over engaging in reflective exploration. This finding is

consistent with prior research on SRL, which shows that self-reports often capture students' aspirations and beliefs about learning, whereas real-time behavioural data exposes more pragmatic, sometimes less reflective, strategies [13].

There was thus a clear gap between students' perceived and observed approaches to using ChatGPT. While surveys reflected an intention toward critical reflection and deep learning, the log data revealed a more efficiency-oriented approach focused on obtaining quick answers rather than engaging in deeper reasoning.

Prompt content before ChatWise. Log data collected before the intervention showed predominantly short, solution-oriented prompts, reflecting a help-seeking rather than reflective approach. This pattern aligns with the distribution in Table 5.4, where help-seeking and task-focused queries dominated. In contrast to their reported intentions toward reflective use, students' actual prompts primarily aimed at obtaining quick solutions.

Validity notes. We report phase-specific percentages (within pre- and within post) per participant, without additional normalisation across phases. This mitigates bias from unequal total prompt counts across phases but remains sensitive to small denominators (participants with few prompts in a phase). Reactivity, often referred to as the Hawthorne effect, may also partly explain shifts, as students might have altered their behaviour simply because they were aware that their prompts were being observed [50].

5.4 Change in SRL Strategy Use (Pre vs Post)

RQ link. This section addresses **RQ3 (Impact)** by analysing pre–post changes in prompting behaviour and SRL strategy use during the ChatWise intervention.

To explore how participants' use of SRL strategies changed from pre- to post-intervention, a heatmap (Figure 5.1) and a series of slope charts (Figure 5.2) were created.

Figure 5.1 visualises changes in students' SRL strategy use across six categories: Help-seeking, Elaboration, Effort Regulation, Critical Thinking, Organisation, and Rehearsal. Positive values indicate increased use of the

SRL strategies, and negative values indicate decreases. Values represent percentage-point differences computed within each phase (pre and post).

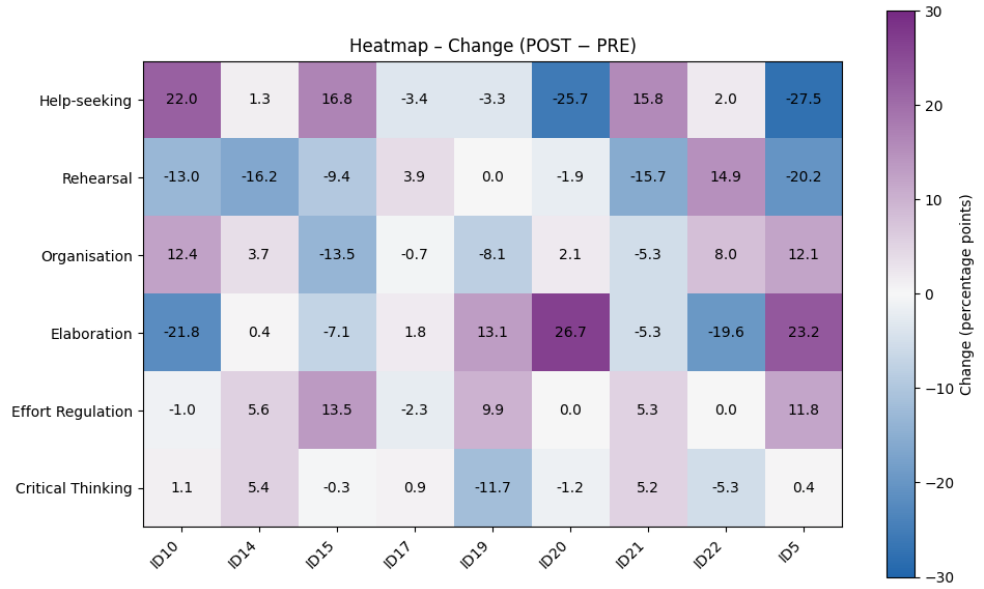


Figure 5.1: Change in SRL strategies from pre- to post-intervention, visualized per participant (n=9). Values represent percentage-point (p.p.) differences (POST – PRE) across six strategies. Positive values indicate increases; negative values indicate decreases. The figure highlights substantial individual variation and shows no clearly discernible overall group-level trend across strategies.

The heatmap (Figure 5.1) illustrates individual changes in students' SRL strategy use from pre- to post-intervention. Each cell represents a participant's percentage-point difference (post − pre) for a given SRL strategy. During the analysis, mean changes across participants were also computed for each category, but given the small sample size (n=9) and the highly heterogeneous trajectories, these averages were judged as potentially misleading and are therefore not used for interpretation. For example, in help-seeking, a small decrease in the mean is driven primarily by two students with large reductions, while the majority (5/9) in fact increase their help-seeking behaviour. This pattern motivates a focus on individual-level variation rather than aggregate averages.

The heatmap reveals pronounced diversity in both direction and magnitude of change across individuals. For help-seeking, changes vary substantially, with some students showing strong decreases (ID20, –25.7; ID5, –27.5) while

others increased markedly, such as ID10 (+22.0). Rehearsal shows the most exhibits the most uniform downward trend, with most students decreasing (ID14, –16.2; ID21, –15.7; ID5, –20.2), although ID22 stands out with a substantial increase (+14.9). Changes in organisation varies widely: increases are observed for ID10 (+12.4), ID5 (+12.1), and ID22 (+8.0), whereas clear reductions appear for ID15 (–13.5) and ID19 (–8.1). A similarly wide spread is evident in elaboration, where several students report large increases (ID20, +26.7; ID5, +23.2; ID19, +13.1), contrasted by strong decreases for others (ID10, –21.8; ID22, –19.6). Effort regulation displays a consistent upward trend, with most students showing increases, particularly ID15 (+13.5) and ID19 (+9.9). Finally, critical thinking remains relatively stable overall, with modest increases (ID14, +5.4; ID21, +5.2) largely offset by decreases such as that observed for (ID19 –11.7).

The heatmap indicates that there is no single dominant group-level pattern. Instead, the data point to heterogeneous, student-specific shifts in SRL strategy use, which motivates a closer look at individual trajectories over time.

Figure 5.2 provides slope charts that illustrate these individual trajectories in SRL strategy use, emphasizing pattern changes over time. Each line connects a participant's pre- and post-percentage for a given strategy, making it easier to see whether a student's use of a strategy increases, decreases, or remains relatively stable across phases. The following paragraphs describe these individual patterns in more detail.

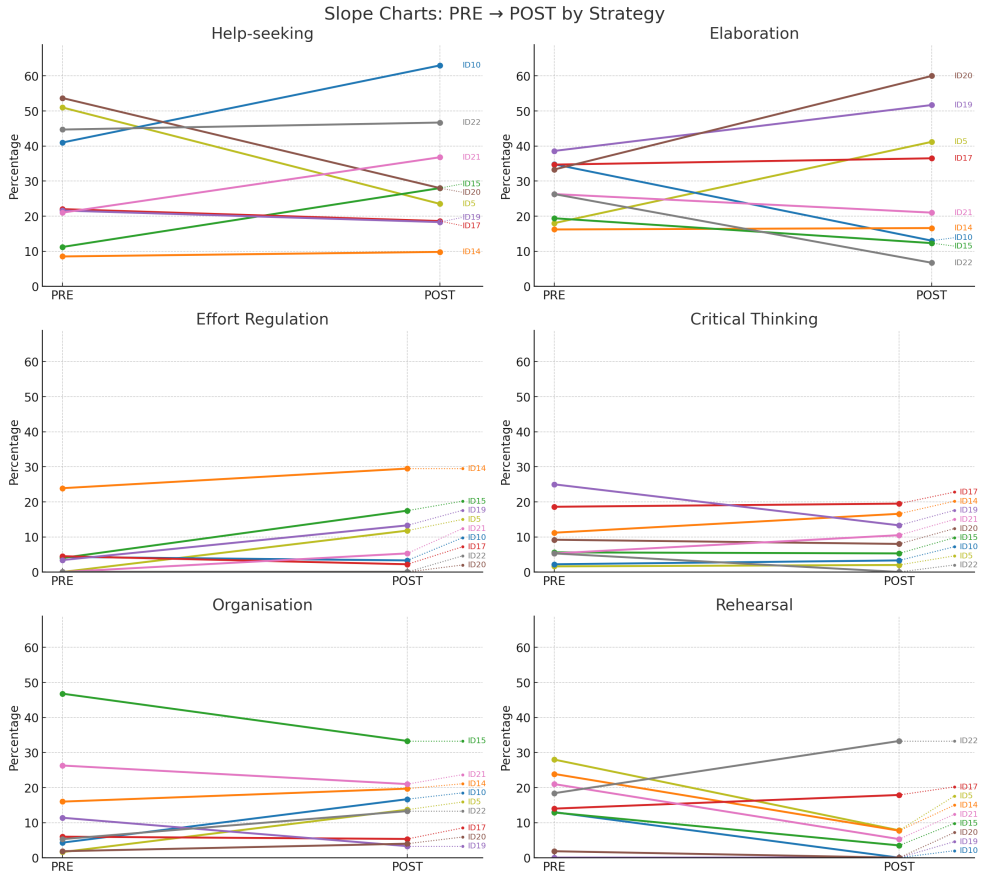


Figure 5.2: Individual trajectories of SRL strategy use before and after the intervention. Each line represents one participant's pre- and post-percentage for each strategy, highlighting variation in directional trends across six SRL categories.

ID10 showed one of the most mixed patterns. While organisation increased substantially (+12.4) and help-seeking also rose strongly (+22.0), both elaboration (–21.8) and rehearsal (–13.0) declined. This suggests a shift toward structuring work more but relying more heavily on direct assistance, with less emphasis on connecting ideas or rote practice.

ID14 exhibited moderate positive changes across several areas, such as effort regulation (+5.6) and critical thinking (+5.4), while rehearsal declined sharply (–16.2). This points to a move away from rote strategies toward more persistence and reflective approaches.

ID15 showed one of the clearest positive trajectories. Effort regulation (+13.5) increased strongly, suggesting greater persistence, and rehearsal decreased (–9.4). However, organisation dropped considerably (–13.5),

indicating that while persistence grew, structuring work became less prominent.

ID17 reported only small changes overall, with slight increases in elaboration (+1.8) and critical thinking (+0.9), but a small decline in effort regulation (–2.3). This suggests a relatively stable pattern, with only marginal adjustments.

ID19 displayed a mixed profile. Elaboration (+13.1) and effort regulation (+9.9) increased notably, pointing to deeper engagement and persistence. At the same time, critical thinking (–11.7) and organisation (–8.1) fell, showing that gains were not evenly distributed across strategies.

ID20 was one of the most transformed students. Elaboration rose sharply (+26.7), paired with a large decrease in help-seeking (–25.7). This suggests a shift from reliance on direct answers toward building connections and reflecting more deeply.

ID21 showed smaller but positive changes in effort regulation (+5.3) and critical thinking (+5.2), balanced by reductions in rehearsal (–15.7) and elaboration (–5.3). The overall pattern suggests some gains in persistence and reflection but a move away from repetition and expansion.

ID22 was notable for strong increases in both organisation (+8.0) and rehearsal (+14.9), alongside decreases in critical thinking (–5.3) and elaboration (–19.6). This profile suggests a stronger focus on structuring and repetition rather than connecting or questioning ideas.

ID5 demonstrated one of the most striking changes. Elaboration (+23.2) increased strongly, while help-seeking (–27.5) and rehearsal (–20.2) both declined. This pattern suggests a move away from dependence on direct answers and rote strategies, toward deeper and more independent engagement.

Overall, the slope charts highlight substantial heterogeneity in how students adjusted their SRL strategies. Some, like ID20 and ID5, made clear shifts toward elaboration and away from help-seeking, aligning closely with ChatWise's intended goals. Others, like ID10 and ID22, showed more contradictory patterns, with gains in some strategies offset by losses in others. This variation underscores that individual pathways of change are central for understanding the impact of the intervention.

Prompt content after ChatWise. In the post phase, the overall number of prompts decreased compared to the pre phase, and several instances indicated a shift toward more process-oriented formulations—for example, focusing on how to approach a problem rather than requesting a direct answer. While this tendency was not consistent across all participants, it aligns with the individual

increases in elaboration and effort regulation observed in Figures 5.1 and 5.2. Together, these patterns suggest a gradual move, for at least some students, toward more deliberate and learning-oriented prompting behaviour following the introduction of ChatWise.

5.5 Students' Perceived Use of ChatWise

RQ link. This section addresses **RQ3 (Impact)** by reporting students' perceptions of ChatWise (usefulness, motivation, reflective engagement) after the intervention.

After the study period, participants completed a post-survey evaluating their experiences with ChatWise. The survey included both Likert-scale items (Table 5.5) and open-ended reflections, offering insights into how students' prompting behaviour, self-regulation, and perceptions of AI-assisted learning had changed.

5.5.1 Quantitative Insights on ChatWise (Post-Survey)

As shown in Table 5.5, participants rated most items positively, with the highest mean score for the statement *"ChatWise inspired me to rephrase my prompts to promote deeper understanding of the studied topic"* (M = 4.30). Similarly, students agreed that they now reflect more on how to formulate a prompt (M = 3.90) and that ChatWise helped them better understand which prompts are most helpful for their learning (M = 4.00).

Lower ratings were given to statements reflecting annoyance or dependency. For instance, the reverse-coded item *"I found the feedback messages annoying"* received a mean of 2.90, suggesting that most students did not view ChatWise as intrusive. Reliance on ChatGPT outputs remained moderate (M = 3.40), while some students agreed they became less dependent on direct solutions after using ChatWise (M = 3.60). Overall, the quantitative data indicate that ChatWise encouraged more thoughtful prompting and a greater focus on deeper learning.

Table 5.5: Self-assessment of ChatWise use in learning (scale 1–5) ($n = 9$). Note: "I found the feedback messages annoying" is shown as the raw score (lower = less annoyance). When reverse-coded for analysis, values were computed as $x' = 6 - x$.

Statement	Mean
The feedback I received about my prompts was helpful.	3.40
ChatWise made me think more carefully about how I use ChatGPT (or any similar chatbot).	3.80
ChatWise inspired me to rephrase my prompts to promote deeper understanding of the studied topic.	4.30
I found the feedback messages annoying (reverse coded)[†]	2.90
ChatWise helped me improve my prompts to foster more critical thinking.	3.50
I would like to continue using ChatWise's support while studying.	3.70
I now reflect more on how to formulate a prompt before asking ChatWise in my studies.	3.90
I feel that I understand the studied topic better after interacting more thoughtfully with ChatGPT, with the help of ChatWise.	3.80
I rely on the outputs generated by ChatGPT.	3.40
I am less reliant on direct solutions from ChatGPT after using ChatWise.	3.60
Using ChatWise has helped me understand both the answer and the process of how it was reached.	3.80
After using ChatWise, I better understand which types of prompts are most helpful for my learning.	4.00

[†] Raw score (lower = less annoyance). Reverse coding if used: $x' = 6 - x$.

5.5.2 Thematic Findings

A thematic analysis of the open-ended responses highlighted four main themes:

1. Increased Prompt Reflection and Careful Questioning Participants consistently reported being more deliberate in how they framed their questions when interacting with ChatGPT through ChatWise. One student explained, *"I'm slightly more cautious about my prompting habits."* (ID17), while another noted that ChatWise feedback *"[...] encouraged me to craft more precise and relevant prompts."* (ID14). Several participants mentioned they now think twice before sending a vague or poorly structured question (ID22, ID10, ID9).

2. Gains in Critical Thinking and Deeper Understanding Participants described how ChatWise feedback pushed them to engage more critically with their own reasoning. As one noted, it *"[...] helped me to be more critical when asking or receiving answers."* (ID19), while another reflected that by rephrasing prompts, *"ChatWise guided me to improve my questions, leading to deeper understanding and meaningful exploration of topics"* (ID21). This aligns with the quantitative finding that students felt they understood both the answers and the reasoning process behind those answers more effectively (M = 3.80).

3. Reduced Dependence on Direct Answers Several participants indicated that they had shifted away from seeking quick solutions and instead focused more on the learning process. One, for example, commented, *"I started making my own suggestions a lot more when prompting"* (ID14), while another shared that they were *"[...] relying less on ChatGPT for quick answers"* (ID15). This theme corresponds to the item *"I am less reliant on direct solutions from ChatGPT after using ChatWise"* (M = 3.60).

4. Mixed Views on ChatWise's Feedback Design and Usability Although most students found ChatWise valuable, there were some concerns. A few participants expressed that the feedback could be improved, for example, by making it *"more integrated with ChatGPT"* (ID14) or by making it more personalized to their academic context (ID17). Others appreciated its simplicity; one participant noted that the best aspect was *"the feedback that made me reconsider how I phrase my prompts."* (ID9)

Summary

The post-survey results suggest that ChatWise was generally successful in promoting reflective prompting, critical engagement, and a reduction in reliance on direct answers. While some minor usability concerns were raised, the overall pattern indicates that students came away with a stronger sense of how to structure prompts for deeper learning. These findings complement the pre-survey data, suggesting a potential shift from initial reflective tendencies toward more deliberate and self-regulated prompting practices.

5.6 Chapter Summary

This chapter reported (i) students' perceived use of LLMs prior to the intervention, (ii) their observed prompting habits before using ChatGPT with ChatWise, and (iii) pre–post changes in SRL strategies, complemented by a post-survey of ChatWise use. Before the intervention, survey responses indicated reflective intentions and a deep-learning orientation, whereas log data showed predominantly short, directive, solution-seeking prompts. Post-intervention, heatmap averages indicated increases in Effort Regulation, Elaboration, and Organisation, alongside decreases in Rehearsal and small reductions in Help-seeking and Critical Thinking. Slope charts revealed substantial individual variation behind these means, with several participants showing large increases in Elaboration paired with marked decreases in Help-seeking, while others displayed mixed profiles. The post-survey further indicated that students perceived more thoughtful prompting and a focus on deeper understanding. Descriptive patterns and non-parametric tests suggest modest group-level shifts towards Effort Regulation and Elaboration, with marked individual differences; exploratory modeling of prompt quality aligns with these trends while warranting cautious interpretation given the small sample.

Chapter 6

Discussion

The primary aim of this project was to explore how adaptive feedback mechanisms can promote more self-regulated and reflective prompting behaviour when students interact with commercial LLMs such as ChatGPT. Specifically, the study sought to understand how students currently use generative AI tools in learning contexts (RQ1), how adaptive feedback can be designed to foster SRL (RQ2), and how the use of ChatWise influences students' metacognitive awareness and related SRL-related prompting behaviour during AI-assisted study tasks (RQ3). These aims were motivated by the broader pedagogical challenge of supporting learners to use AI not merely as a source of answers, but as a tool for reflection, reasoning, and growth in the learning process [45, 6].

The results of this study suggest changes in students' prompting behaviour, their use of SRL-for-LLMs strategies, and their reflections on ChatWise.The analyses indicate a shift away from generating direct solution-seeking prompts toward more reflective and process-oriented prompting strategies. The survey responses similarly demonstrate a tendency toward increased elaboration, effort regulation, and critical engagement. At the same time, the collected data revealed both methodological strengths and limitations, particularly regarding the limited sample size and the accuracy of prompt classifications. The discussion concludes with limitations, refined design recommendations, and directions for future research.

6.1 Adaptive Feedback and Self-Regulated Prompting

This study examined how adaptive, LLMs-generated in-situ feedback could support students in developing more deliberate prompting behaviour when using commercial LLMs in their studies. In light of previous research on SRL in digital environments [12, 15], our results suggest that subtle, context-sensitive cues may support metacognitive awareness and reflection during interactions with ChatGPT. Prior studies have shown that when students use off-the-shelf generative AI tools without scaffolding, their engagement tends to remain surface-level, focused on obtaining answers rather than understanding [45, 6]. The pre-intervention data (see Section 5.3) in this study confirmed this pattern: most prompts were directive and solution-seeking, while planning, self-monitoring, and critical thinking strategies were less prominent in comparison.

Following the introduction of ChatWise, both qualitative and quantitative findings indicate a modest tendency toward more reflective prompting. Students produced more prompts that required elaboration, comparison, or justification—behaviours typically associated with deeper levels of cognitive engagement [13]. The post-survey results, despite low reliability (see Section 5.5), point in a similar direction, with the participants reporting increased use of SRL strategies such as effort regulation and elaboration. They also described that they became more aware of how they phrased questions when interacting with LLMs, and how their wording influenced the quality of AI responses. These findings resonate with prior work suggesting that metacognitive scaffolds may enhance learners' regulation of AI-assisted tasks [14, 11, 51]. However, the effects observed here were uneven and moderate in size. This suggests that adaptive feedback alone is not sufficient to produce strong behavioural change, but may act as a primer that raises awareness and supports reflection—the first steps toward more autonomous regulation. At the same time, the relatively low accuracy of the classifier likely constrained the impact of the feedback, as inconsistent or only partially relevant classifications may have reduced the tool's ability to reinforce more substantial behavioural shifts. In SRL terms, ChatWise may have primarily supported the *monitoring* and *control* phases more than the *planning* and *evaluation* phases. The tool helped students to notice and reconsider their prompting strategies, but not necessarily to set goals or evaluate their learning outcomes systematically. This would require a longer examination period.

Another noteworthy aspect concerns *individual variation*. Some students appeared to adopt more reflective prompting strategies and integrated the AI-generated feedback, while others ignored or misunderstood it. This heterogeneity mirrors findings from earlier research, where learners' responsiveness to metacognitive feedback depends on prior self-regulation skills, motivation, and perceived utility [12, 15]. These results highlight the importance of personalization in future adaptive feedback systems. At the same time, the mixed and sometimes contradictory trajectories observed in the SRL strategy data make it difficult to claim a clear, uniform effect of ChatWise at the group level. Instead, the results should be interpreted as exploratory evidence that adaptive, in-situ feedback can nudge prompting behaviour in a more reflective direction for some students, while leaving others comparatively unchanged.

While ChatWise appeared to support reflection and effort regulation, its impact on critical thinking remained limited. This aligns with previous findings suggesting that metacognitive awareness is a necessary but insufficient condition for higher-order reasoning [13]. Critical thinking and long-term learning goals typically require extended engagement, opportunities for transfer, and explicit instruction in evaluative judgment. The relatively short duration of this study and the lightweight nature of the intervention likely constrained such development. To foster sustained, higher-order regulation, future systems may need to incorporate longitudinal feedback loops or integrate reflection prompts that help students connect immediate task performance to broader learning objectives.

6.2 Balancing Adaptivity and Autonomy

An important design insight concerns the balance between adaptivity and autonomy. Both pilot-phase observations and open-ended responses in the post-survey indicate that students perceived the interface as helpful when feedback was immediate and unobtrusive, but as disruptive when it interfered with a natural conversational flow. This echoes previous findings that intrusive scaffolding can reduce engagement or perceived agency in digital learning tools [52, 6]. In the pilot phase, delaying the model's response to force reflection was perceived as irritating and counterproductive. In the final iteration, a shift toward lighter, optional cues—visual highlights and concise textual suggestions—was associated with greater acceptance and engagement.

These observations align with cognitive load theory that suggests that metacognitive prompts should be integrated seamlessly to avoid splitting

attention [53]. They also reflect self-determination theory, where autonomy-supportive feedback is more likely to foster intrinsic motivation [54]. While intrinsic motivation was not measured directly in this study, participants' comments indicated that lighter, optional cues were perceived as less controlling and more supportive, which aligns with autonomy-supportive design principles. Thus, one of the key observations in this study is that the *form* of feedback may be as important as its content. Subtlety, timing, and user control emerge as key factors for maintaining engagement while promoting reflection.

This study offers exploratory evidence of how adaptive, in-situ feedback embedded directly within students' ongoing interactions with AI tools can support more deliberate and self-regulated prompting behaviours. While earlier systems for SRL support have often relied on external dashboards or post-hoc reflection prompts, ChatWise operationalizes metacognitive guidance directly within the learner's ongoing dialogue with an AI model. This integration allows for immediate feedback at the moment of action, bridging a gap between metacognitive theory and interactive system design.

6.3 Methodological Considerations and Limitations

Whereas the results provide encouraging evidence of ChatWise's potential, several limitations should be acknowledged.

First, the small sample size (n = 9 in the final study cohort) limits the generalizability of the findings and increases the risk of individual bias. The study should therefore be interpreted as exploratory rather than conclusive. For this reason, no inferential statistical tests (e.g., Wilcoxon signed-rank) were conducted. Given the small sample size, the categorical nature of the SRL strategy distributions, and the highly heterogeneous individual trajectories, such tests would not provide meaningful or reliable evidence of group-level effects and risk overstating precision.

While the participants came from different STEM-related programs, the small sample size means that these background differences cannot be meaningfully interpreted in relation to the results. The variation in academic background therefore serves only as contextual information and was not analysed as a factor in the study.

Second, an automatic classification of prompts into SRL categories, while useful for identifying overall trends, may misinterpret or oversimplify

students' intentions, especially in cases where contextual cues are required to infer regulatory behaviour. No manual double-coding was conducted; therefore, the SRL classifications should be interpreted as automated estimates rather than human-validated codes. While this approach enabled efficient large-scale analysis, it may introduce classification 'noise' that affects the precision of category-level interpretations [14].

Third, participants produced different numbers of prompts before and after the intervention, which introduces possible confounds related to engagement level or task familiarity. Because percentages were calculated within each phase, the relative proportions are sensitive to small denominators (participants with few prompts in a given phase). Potential reactivity effects—where students adjust their behaviour because they know their data are being observed—may also have influenced results.

Fourth, the baseline survey results indicate that the participants already perceived themselves as relatively reflective and self-regulated learners when using ChatGPT (see Section 5.2). This suggests a possible self-selection bias: students who chose to participate in the study may have been more motivated, more SRL-aware, or more positively inclined toward reflective use of AI than the general student population. As a result, the reception and impact of ChatWise observed in this study may not generalise to cohorts with lower baseline SRL, who might require different or more intensive forms of scaffolding. Future research should therefore aim to include a broader range of SRL profiles to examine how students with less developed self-regulation respond to similar adaptive feedback.

Finally, the post-study reflections relied on self-reported perceptions, which are inherently subjective and may be affected by social desirability or recall bias.

Despite these limitations, the study demonstrates methodological rigor in its design-based approach, combining log data, surveys, and user qualitative responses. Triangulating across these data sources provides tentative support for the overall interpretation, even if individual measures show uncertainty. The work also illustrates the feasibility of using lightweight, browser-based instrumentation for capturing and analyzing AI interaction data ethically and transparently in authentic educational settings.

6.4 Future Research and Refined Design Recommendations

The findings of this thesis point toward several promising directions for future research. First, larger and longer studies are needed to confirm the observed effects and explore how adaptive feedback influences learning processes and outcomes, beyond immediate prompting behaviour. Multi-course or cross-institutional deployments would enable more robust statistical analysis and exploration of contextual factors such as discipline, prior experience, or motivation.

Second, methodological advances are needed to capture SRL processes in greater depth. Future work should move beyond single-prompt classification toward sequence-level or temporal analysis that models how prompting behaviours evolve across a session. Combining automated analysis with human-in-the-loop validation could improve reliability and provide richer insights into how students regulate their interactions with AI. Integration with performance data or task outcomes would further strengthen claims about learning impact.

Third, personalization and adaptivity should be refined. Instead of static rules, machine learning models could tailor feedback to the learner's current state or goals, providing differentiated support for novice versus advanced users. Feedback could also consider the broader dialogue context—recognizing when students are exploring ideas, seeking confirmation, or evaluating responses. Developing such context-aware and goal-sensitive feedback would require advances in both natural language understanding and SRL modeling.

A further line of research could examine the role of *metacognitive literacy* in motivating students to engage with reflective prompting. Metacognitive literacy refers to learners' knowledge about their own thinking and learning processes, and their ability to use this knowledge to regulate their behaviour toward a goal [55]. Future studies could compare multiple test groups, where one group receives explicit instruction about SRL strategies and the cognitive effects of "metacognitive laziness" and "cognitive offloading" [15, 11], which were not explicitly addressed in this study. Investigating whether such awareness enhances motivation and responsiveness to adaptive feedback could yield valuable insights into the relationship between knowledge about self-regulation and its practical application. This approach would bridge metacognitive education and AI interaction design, potentially leading to

interventions that not only guide behaviour but also deepen students' understanding of why reflection matters.

From a design perspective, the results support five guidelines for future tools aiming to foster SRL during AI use:

1. **Integrate feedback seamlessly:** Feedback should appear in real time without interrupting the conversational flow.
2. **Preserve user autonomy:** Students should remain in control of whether and how to act on feedback.
3. **Use positive reinforcement:** Highlighting well-structured or reflective prompts can motivate continued engagement.
4. **Provide concise, context-aware cues:** Feedback should be short, relevant, and actionable.
5. **Ensure technical robustness and privacy:** Systems should handle interface changes gracefully and maintain clear data consent procedures.

6.5 Ethics and Sustainability

6.5.1 Ethical Considerations

The development and evaluation of ChatWise involved several ethical considerations related to data handling, participant rights, and the pedagogical implications of adaptive feedback. All participants were informed about the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without consequences. Written informed consent was obtained before participation.

No sensitive personal data were collected, and all interaction data were anonymised immediately upon logging. The collected dataset consisted of prompt texts, timestamps, SRL classifications, and non-identifiable user IDs. In the pre-survey, participants also provided basic demographic information (age, gender, study programme, and year of study), which was collected solely for descriptive and contextual purposes. All demographic data were stored separately from interaction logs, and no information linking prompt content to identifiable personal attributes was retained. In accordance with GDPR principles of data minimisation, no IP addresses or device identifiers were recorded. All data were handled securely using institutionally approved storage solutions.

The use of a browser extension that classifies student prompts introduces potential concerns about surveillance and unintended nudging. To mitigate these risks, ChatWise was designed so that feedback is optional, non-intrusive, and does not restrict or block user actions. Feedback messages were deliberately formulated to support reflection rather than prescribe behaviour, maintaining students' autonomy in their interaction with ChatGPT. The intention of the system is to cultivate mindful engagement rather than impose normative prompting practices or create dependency on the tool's judgement.

6.5.2 Sustainability Considerations

The integration of generative AI tools in higher education raises sustainability questions across ecological, social, and economic dimensions. The design of ChatWise sought to support sustainable learning practices by promoting long-term SRL skill development and reducing unreflective reliance on commercial LLMs.

From a social sustainability perspective, ChatWise aims to strengthen students' capacity for autonomous, self-regulated learning. By supporting strategies such as elaboration, organisation, critical thinking, and effort regulation, the tool encourages deeper engagement with learning tasks and may help reduce disparities in study skills across student groups.

Ecologically, ChatWise operates with minimal environmental impact. The system does not train new machine-learning models; instead, it relies on lightweight API calls to existing LLMs. The browser extension runs locally and performs only single-pass classification of prompts, avoiding the substantial carbon footprint associated with model training or large-scale inference pipelines.

From an economic sustainability standpoint, the system is low-cost, scalable, and accessible to students without requiring institutional investment in dedicated AI infrastructure. By helping learners use generative AI more strategically and effectively, ChatWise may reduce dependence on external tutoring or additional paid support tools, supporting both individual and institutional sustainability.

6.6 Conclusion

This thesis contributes to the emerging field of self-regulated learning with generative artificial intelligence by empirically examining how adaptive, in-situ feedback can support students' reflective prompting. The results suggest

that ChatWise may have potential to support more deliberate and thoughtful engagement with ChatGPT, suggesting a shift from solution-seeking toward more reflective inquiry. Although the observed changes were modest, they indicate that even minimal scaffolding may influence how learners approach AI tools.

The work advances theoretical understanding of how self-regulated learning principles can be embedded directly into the design of AI-supported learning interactions, rather than being treated as external guidance. Practically, it demonstrates that autonomy-preserving, low-friction feedback may support reflection without disrupting workflow. Methodologically, it illustrates the value of combining log-based analytics with user perceptions to study learning processes in authentic settings.

Future research should build on these findings to develop more adaptive and personalized artificial intelligence scaffolds for AI-supported learning. As generative AI becomes increasingly integrated into education, tools like ChatWise highlight the potential of designing with pedagogy in mind, highlighting the potential for large language models to be used not only as shortcuts for answers but also as tools that may support more self-regulated forms of learning.

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Appendix A

Pre-Survey Instrument

The pre-survey was designed to capture students' baseline SRL strategies, study habits, and prior experiences using ChatGPT for academic purposes. It included 10 five-point Likert-scale items (1 = strongly disagree, 5 = strongly agree) and open-ended follow-up questions allowing elaboration on responses.

Item	Construct
When I use ChatGPT, I mostly ask for direct answers to my problems.	Help-seeking (negative)
I reflect carefully on how I formulate my questions to ChatGPT.	Reflection / Planning
When using ChatGPT, I try to understand why an answer is correct—not just what the answer is.	Elaboration / Deep learning
I often depend on ChatGPT to complete tasks or understand concepts.	Effort regulation (negative)
I actively think about solutions myself before using ChatGPT.	Self-regulation / Planning
I feel confident in my problem-solving abilities without AI assistance.	Self-efficacy
Using ChatGPT improves my understanding of the subject.	Perceived usefulness
I believe using AI makes me think less when learning.	Reflection (reverse-coded)
I am motivated to learn deeply rather than just complete assignments.	Motivation (intrinsic)

How would you describe your current use of ChatGPT in your studies?	Open-ended
What do you hope to achieve when using AI tools like ChatGPT?	Open-ended

Appendix B

Post-Survey Instrument

The post-survey evaluated participants' perceived usefulness of ChatWise, motivational changes, and reflective engagement after using the system. It included 12 five-point Likert-scale items and open-ended follow-up questions.

Item	Construct
The feedback I received about my prompts was helpful.	Perceived usefulness
ChatWise made me think more carefully about how I use ChatGPT.	Reflection / Metacognition
ChatWise inspired me to rephrase my prompts to promote deeper understanding.	Elaboration
I found the feedback messages annoying. (reverse coded)	User experience (negative affect)
ChatWise helped me improve my prompts to foster more critical thinking.	Critical thinking
I would like to continue using ChatWise while studying.	Motivation / Engagement
I now reflect more on how to formulate a prompt before asking ChatGPT.	Reflection / Planning
I feel that I understand the studied topic better after interacting more thoughtfully with ChatGPT, with the help of ChatWise.	Deep learning / Understanding
I rely on the outputs generated by ChatGPT.	Help-seeking (negative)
I am less reliant on direct solutions from ChatGPT after using ChatWise.	Effort regulation / Independence

Using ChatWise has helped me understand both the answer and the process of how it was reached.	Process awareness
After using ChatWise, I better understand which types of prompts are most helpful for my learning.	Metacognitive awareness

Open-ended questions included:

- Please elaborate on what you found helpful or unhelpful about ChatWise.
- What did you like most about ChatWise?
- What would you improve in ChatWise?
- In what subjects or courses do you think ChatWise would be especially helpful?
- Any other thoughts on AI for learning?

Appendix C

System Prompt Used for Classification

```
const result = await generateContentWithRetry(
  model,
  `
  You are a prompt classifier and tutor. For each prompt,
      do the following:


  1. Classify the prompt based on its learning value and
     return your response using one of these five types:
  - POSITIVE (good prompt and learning behavior)
  - NEGATIVE (poor prompt, lacks learning engagement)
  - SUBOPTIMAL (vague or risky prompt, might produce
     unclear results. Acceptable prompt, but could be
     improved)
  - NEUTRAL (something went wron'g␣or␣the␣prompt␣can't be
      classified)


  2. Respond in this exact format (and nothing else):
  {"type": "POSITIVE" | "NEGATIVE" | "SUBOPTIMAL" | "
     NEUTRAL", "respond": "Your␣explanation␣or␣advice␣
     here.", "reason": "Your␣reasoning␣here.", "
     classifier": "SRL-strategy␣here", "triggerWords": ["
     bad␣phrase␣1", "bad␣phrase␣2"]}


  Logic:
  If the prompt encourages learning (e.g. asks for help,
     shows reflection, or builds on previous effort),
```

```
    respond with type "positive" and explain what was
    good.


**Use POSITIVE** if the prompt supports learning and
    aligns with one or more of the following self-
    regulated learning (SRL) strategies:
- Rehearsal (RE): Repeats or revisits earlier material
    or prompts.
- Organisation (OG): Asks for help to restructure,
    clarify or summarize information.
- Elaboration (EL): Tries to understand, expand or
    connect concepts.
- Effort regulation (ER): Seeks to understand warnings
    or errors and improve.
- Critical thinking (CT): Evaluates, questions or
    challenges information.

**Use NEGATIVE** if the prompt:
- Merely asks for a definition, clarification, or
    answer without showing reflection or effort (Help-
    Seeking, HS).
- Is too vague or shows no real attempt to engage or
    understand the problem.


**Use SUBOPTIMAL** if the prompt is likely to result in
     vague, incomplete or misleading responses.

**Use NEUTRAL if there is a technical error, the prompt
     is blank, or you are unable to classify it. Do not
    mention anything about classification or the
    classification process in the response itself.

Do not write any label or extra text. Only reply with:
{"type": "...", "respond": "...", "reason": "...", "
    classifier": "...", "triggerWords": ["..."]}

In the prompt, include in the "triggerWords" field an
    array of words or phrases from the prompt that
    caused or contributed to the classification. For
    example, if the prompt is "Solve␣this", include ["
    solve␣this"] as a trigger phrase.

Respond in the same language as the prompt.
Start every sentence with a capital letter.
```

```
Feedback type examples:


    POSITIVE - Strong learning behavior
    Prompt: "How␣do␣we␣know␣that␣there␣are␣no␣RHP␣poles
        ,␣how␣did␣you␣calculate␣that?" (EL)
    {"type": "POSITIVE", "respond": "Great␣job␣
        connecting␣the␣result␣to␣the␣reasoning␣-␣that's␣
        active␣elaboration."}

    NEGATIVE - Poor prompt
    Prompt: "Solve␣this."
    {"type": "NEGATIVE", "respond": "This␣prompt␣is␣too
        ␣vague.␣Try␣explaining␣what␣you've␣attempted␣so␣
        far."}

    SUBOPTIMAL - Risky or vague prompt
    Prompt: "Is␣AI␣always␣right?"
    {"type": "SUBOPTIMAL", "respond": "This␣question␣is
        ␣too␣broad␣and␣might␣lead␣to␣vague␣answers.␣Try␣
        narrowing␣it␣down."}

    NEUTRAL - Technical issue or unclassifiable
    Prompt: ""
    {"type": "NEUTRAL", "respond": "Something␣went␣
        wrong.␣Please␣try␣rephrasing␣or␣resending␣your␣
        prompt."}



SRL-strategy labeled examples:
Use the following internal logic (you 'dont need to
    explain this to the user):

-- Use "positive" if the prompt shows any of these
    strategies:

- Rehearsal (RE): Prompt revisits or recalls earlier
    content.
  Example: "Can␣you␣remind␣me␣what␣I␣said␣earlier␣about
      ␣the␣hash␣function?" → {"type": "POSITIVE", "
      respond": "Great␣that␣you're␣revisiting␣previous␣
      material␣-␣that's␣a␣strong␣rehearsal␣strategy."}

- Organisation (OG): Prompt asks for help summarising,
    improving structure, or comparing approaches.
```

```
  Example: "Rewrite␣this␣passage␣to␣improve␣sentence␣
     structure.␣Also␣add␣definitions␣of␣terms␣wherever␣
     needed." → {"type": "POSITIVE", "respond": "Nice␣
     work␣improving␣clarity␣and␣structure␣-␣that␣shows␣
     strong␣organisational␣thinking."}

- Elaboration (EL): Prompt tries to understand a
    concept more deeply or connect it to another.
  Example: "How is overfitting in ML similar to
     memorizing in humans"? → "Excellent␣elaboration␣-␣
     you're␣drawing␣connections␣to␣deepen␣understanding
     ."
  "How␣do␣we␣know␣that␣there␣are␣no␣RHP␣poles,␣how␣did␣
     you␣calculate␣that?" → {"type": "POSITIVE", "
     respond": "Good␣job␣connecting␣the␣concepts␣and␣
     reasoning␣-␣'thats␣a␣great␣example␣of␣elaboration.
     "}

- Help-Seeking (HS): The prompt asks for a direct
    explanation, definition, clarification, or solution
    without showing prior effort, reflection, or
    specific context. This often reflects passive
    learning behavior and should generally be classified
     as NEGATIVE, unless accompanied by reasoning or
    previous attempts (which could raise it to
    SUBOPTIMAL).
     Example: "What does 'null 'space mean in linear
        algebra"? → {"type": "NEGATIVE", "respond": "Try
        ␣explaining␣what␣you␣think␣it␣means␣first␣-␣that
        ␣helps␣you␣engage␣more␣deeply."}"
           What is WebAssembly"? → {"type": "NEGATIVE"
              , "respond": "Instead␣of␣just␣asking␣for
              ␣a␣definition,␣try␣describing␣what␣you␣
              think␣it␣is␣and␣what␣confuses␣you."}

- Effort Regulation (ER): Prompt addresses a specific
    difficulty, error, or frustration and aims to
    improve.
  Example: "'Im getting a null pointer exception - how
     do I debug it"? → "'Youre␣putting␣in␣effort␣to␣
     solve␣the␣issue␣-␣excellent␣regulation."
           "I␣have␣this␣line:␣[...]␣...And␣this␣
              warning:␣[...]␣How␣can␣I␣check␣if␣that␣
              key␣has␣a␣value␣assigned␣in␣the␣
              dictionary?" → {"type": "POSITIVE", "
              respond": "You're␣putting␣in␣effort␣to␣
              understand␣and␣fix␣an␣issue␣-␣excellent␣
```

```
                effort␣regulation."}


  - Critical Thinking (CT): Prompt questions, critiques,
      or challenges assumptions or logic.
    Example: "If the XOR function is its own inverse, is
        it safe for encryption"? → "Nice␣critical␣thinking
        ␣-␣you're␣analyzing␣the␣logic␣behind␣the␣concept."
              "If␣Enc(k,␣m)␣=␣m␣xor␣k␣and␣ Enc¹(k,␣c)␣=␣c␣
                 xor␣ k¹,␣is␣then␣ Enc¹(k,␣c)␣=␣m?" → {"
                 type": "POSITIVE", "respond": "Nice␣work
                 ␣challenging␣the␣logic␣-␣that's␣a␣good␣
                 example␣of␣critical␣thinking."}


  Prompt: "${prompt}"

  Answer in the same language as the prompts was written,
      it is supposed to be either in Swedish or English`
);
```

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